Introduction to MATLAB MTH643

Virtual University of Pakistan

What is MATLAB

MATLAB is a programming and numeric computing platform used by engineers and scientists to analyze data, develop algorithms, and create models.

It is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation.

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Industry Applications



Automated Driving Systems

Design, simulate, and test automated driving systems



Computational Biology

Analyze, visualize, and model biological data and systems



Control Systems

Design, test, and implement control systems



Image Processing and Computer Vision

Acquire, process, and analyze images and video for algorithm development and system design



Internet of Things

Connect embedded devices to the Internet and gain insight from your data



Machine Learning

Train models, tune parameters, and deploy to production or the edge



Data Science

Explore data; build machine learning models; do predictive analytics



Deep Learning

Data preparation, design, simulation, and deployment for deep neural networks



Embedded Systems

Design, code, and verify embedded systems



Enterprise and IT Systems

Use MATLAB with your IT systems



Mechatronics

Design, optimize, and verify mechatronic systems



Mixed-Signal Systems

Analyze, design, and verify analog and mixed-signal systems



Power Electronics Control Design

Design and implement digital control for motors, power converters, and battery systems



Power Systems Analysis and Design

Design and simulate electric grids and transportation systems



Predictive Maintenance

Develop and deploy condition monitoring and predictive maintenance software



Robotics

Design, simulate, and verify robotics and autonomous systems

- Robot Manipulators
- Mobile Robots
- UAV



Signal Processing

Analyze signals and time-series data. Model, design, and simulate signal processing systems



Test and Measurement

Acquire, analyze, and explore data and automate tests



Wireless Communications

Create, design, test, and verify wireless communications systems

Toolboxes

MATLAB features a family of application-specific solutions called toolboxes. Very important to most users of MATLAB, toolboxes allow you to *learn* and *apply* specialized technology. Toolboxes are comprehensive collections of MATLAB functions (M-files) that extend the MATLAB environment to solve

particular classes of problems. Areas in which toolboxes are available include signal processing, control systems, neural networks, fuzzy logic, wavelets, simulation, and many others.

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Course Contents

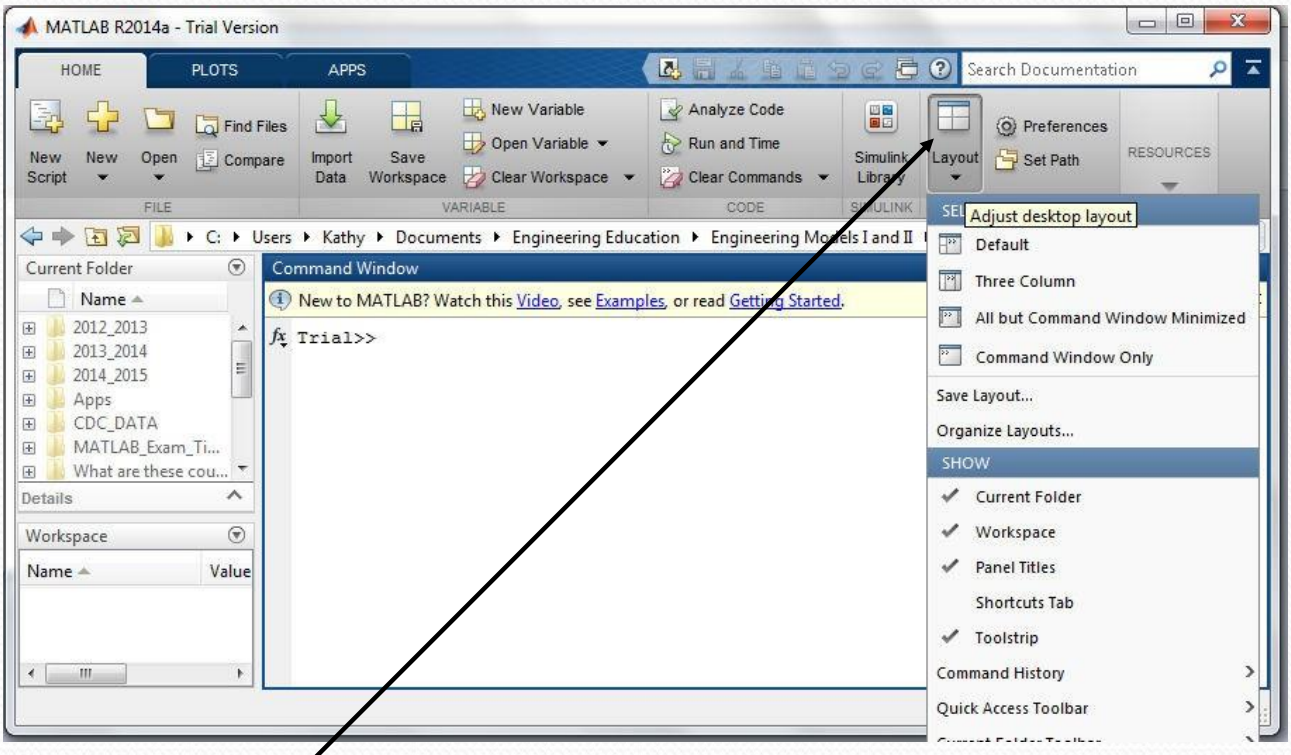
- Variables, and data types
- Script Files
- Conditional Statements
- Loops, Nested Loops
- Arrays
- Functions
- Plotting
- Use of MATLAB to solve Mathematical Problems

MATLAB® Desktop

MATLAB Desktop

- The **Command window** is where you type MATLAB commands following the prompt: >>
- The **Workspace window** shows all the variables you have defined in your current session. Variables can actually be manipulated within the workspace window.
- The **Command History** window displays all the MATLAB commands you have used recently – even includes some past sessions.
- The **Current Folder** window displays all the files in whatever folder you select to be current.

MATLAB Desktop

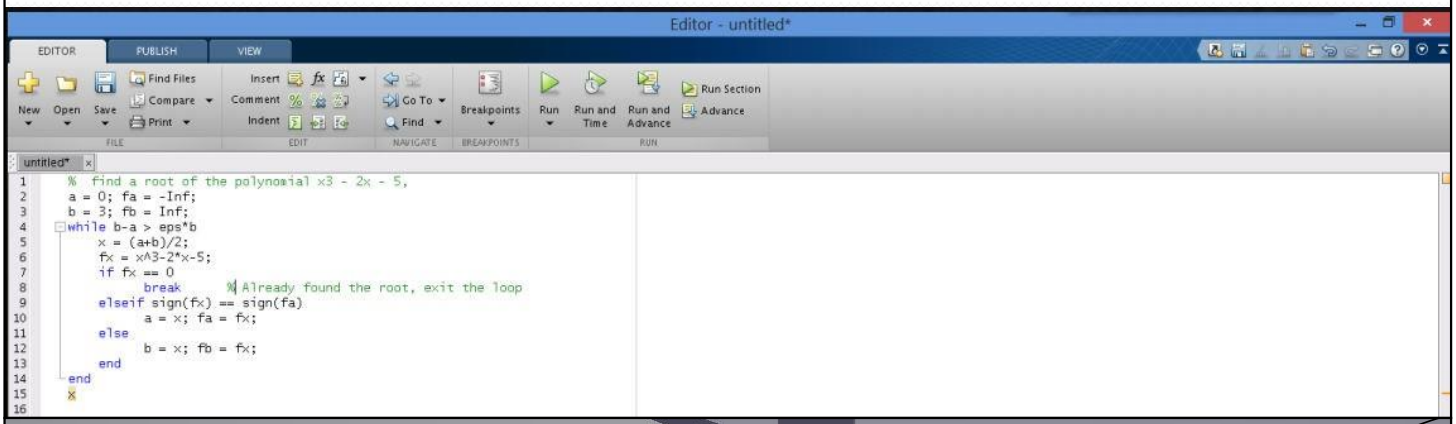


You can select what is on your desktop by Clicking on Layout. Go down to Command History and select docked.

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M-File

- An m-file, or script file, is a simple text file where you can place MATLAB commands.
- Save your work
- Convenient for debugging
- Run directly



Using MATLAB® As a Calculator

Arithmetic Operators and Order of Operations

- Addition (+), Subtraction (-), Multiplication (*), Division (/), Power (^)
- Order of Operations (same rules you should already know from math class and using a calculator)
    1. Complete all calculations inside parenthesis or brackets using the precedent rules below
    2. Powers (left to right)
    3. Multiplication and Division (left to right)
    4. Addition and Subtraction (left to right)

Operator	Purpose
+	Plus; addition operator.
-	Minus; subtraction operator.
*	Scalar and matrix multiplication operator.
.*	Array multiplication operator.
^	Scalar and matrix exponentiation operator.
.^	Array exponentiation operator.
\	Left-division operator.
/	Right-division operator.
.\	Array left-division operator.

./	Array right-division operator.
:	Colon; generates regularly spaced elements and represents an entire row or column.
( )	Parentheses; encloses function arguments and array indices; overrides precedence.
[ ]	Brackets; enclosures array elements.
,	Comma; separates statements and elements in a row
;	Semicolon; separates columns and suppresses display.
%	Percent sign; designates a comment and specifies formatting.
=	Assignment operator.

Arithmetic Operators and Order of Operations

Some Examples:

```
>>  10/5*2
```

```
>>   5*2^3+4(2)

>> -1^4

>>   8^1/3
```

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Exercise 1

Calculate the area and circumference of a circlewith a radius of 4 cm.

Area = 50.27 cm^2

Circumference = 25.13 cm

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1 Doing Mathematics with Matlab (I)

2 Exercise

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1 Doing Mathematics with Matlab (I)
- Arithmetic
- Symbolic Computation
- Substitution in Symbolic Expressions
- Errors in input

2 Exercise

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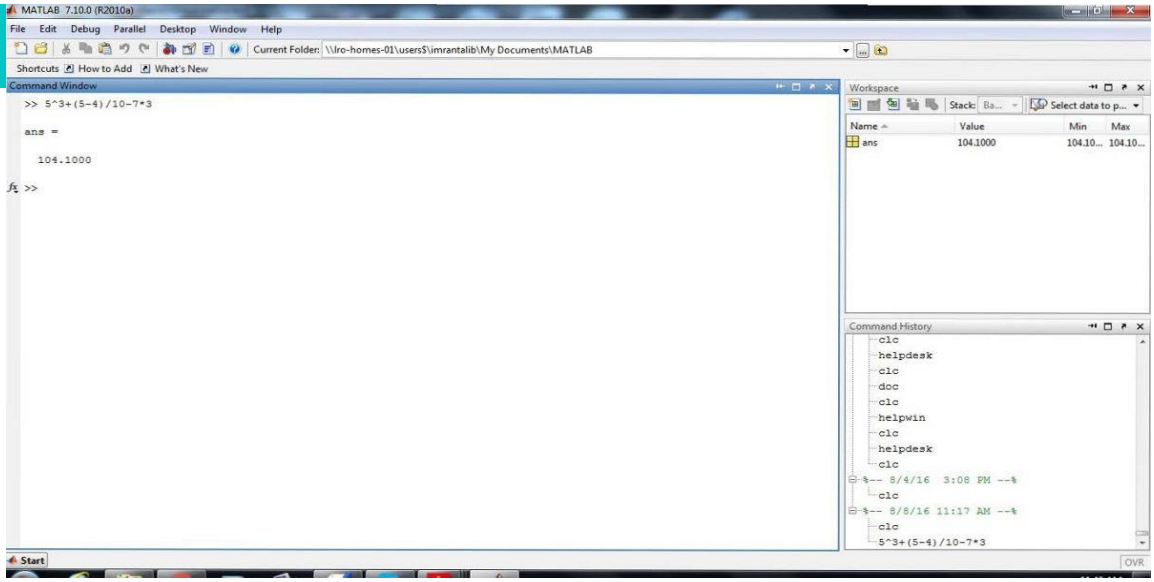


Figure: (1) The Matlab Desktop display after performing some basic arithmetic commands

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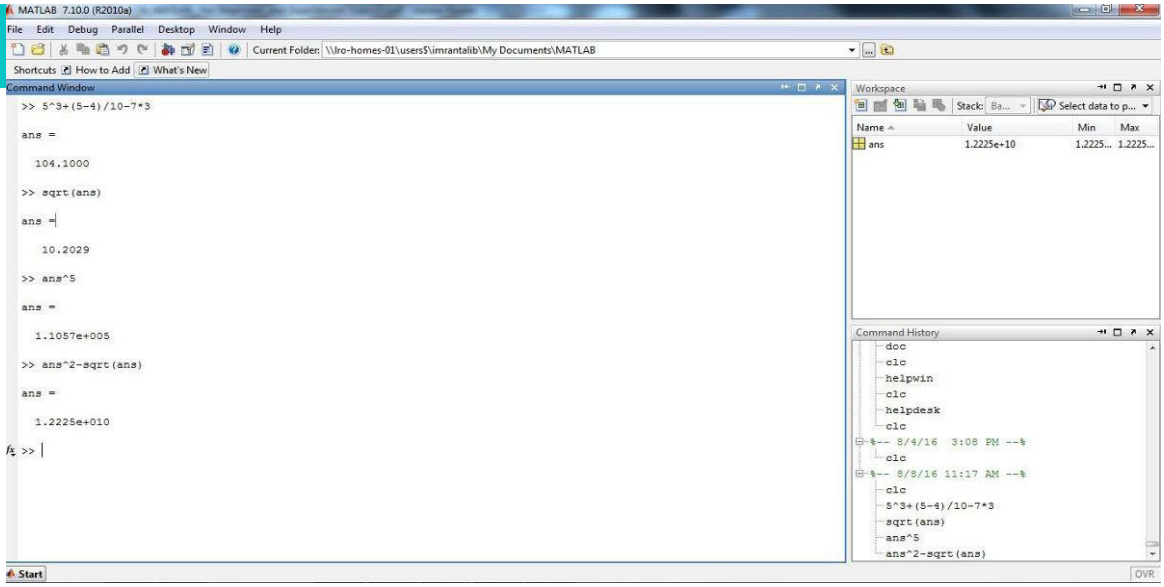


Figure: (2) The Matlab Desktop display after performing some basic arithmetic commands

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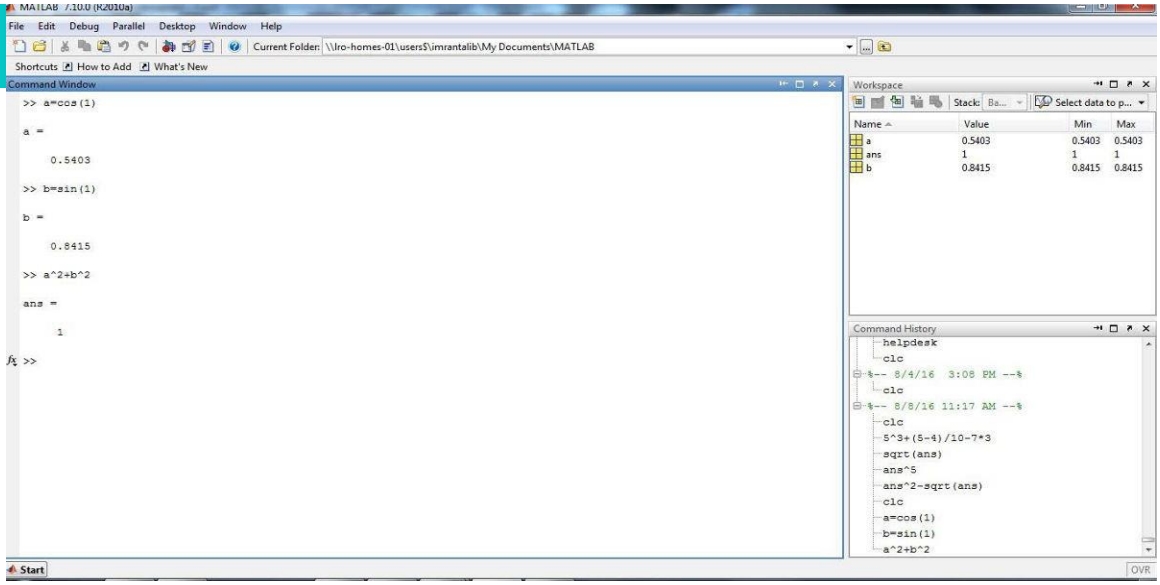


Figure: (3) The Matlab Desktop display after performing some basic arithmetic commands

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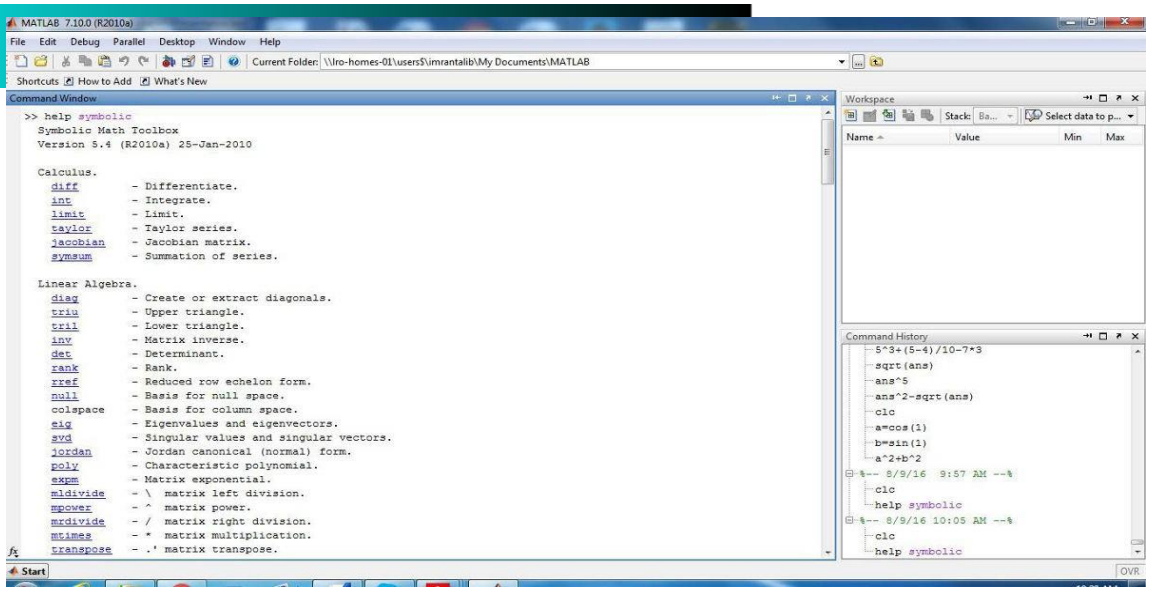


Figure: (4) The Matlab Desktop display with help symbolic command

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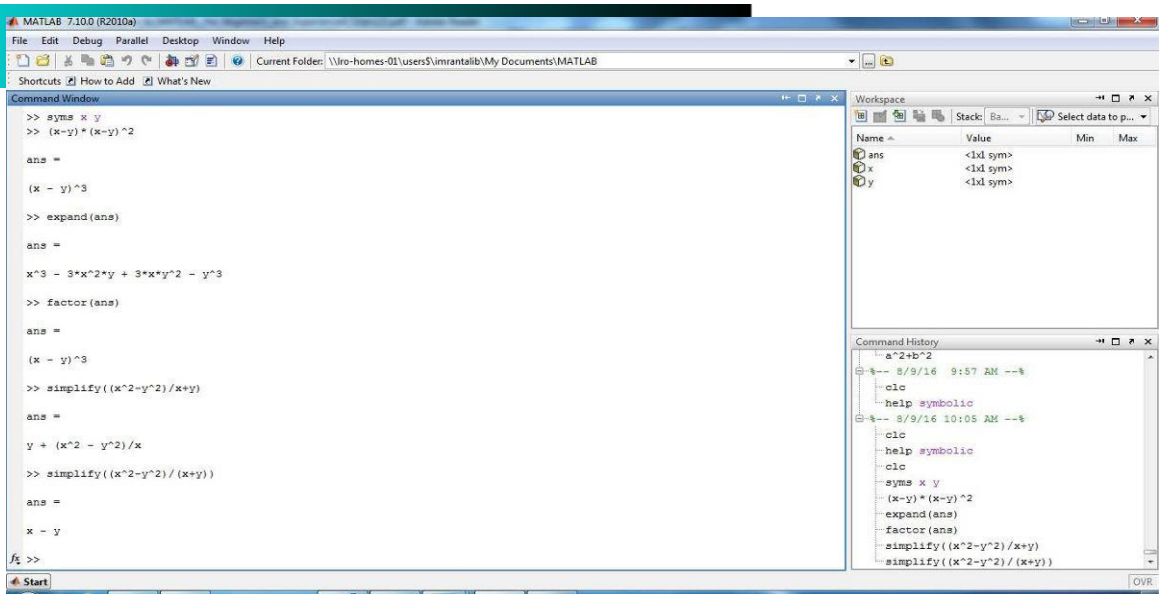


Figure: (5) The Matlab Desktop display after declaring *x* and *y* to be symbolic

It is to be noted that MATLAB uses floating point arithmetic for its calculations. But exact arithmetics can also be done using the Symbolic Math Toolbox with symbolic expressions.

For example, the floating point format of $\cos(pi/2)$ *in MATLAB is* $6.1232e-17$. *However, it is known that* $\cos(pi/2)$ *is really equal to zero. The inaccuracy is due to the fact that typing pi in MATLAB gives an approximation to* π *accurately to about 15 digits. But sym command gives accurate value of* π. *Look at the Figure 6*

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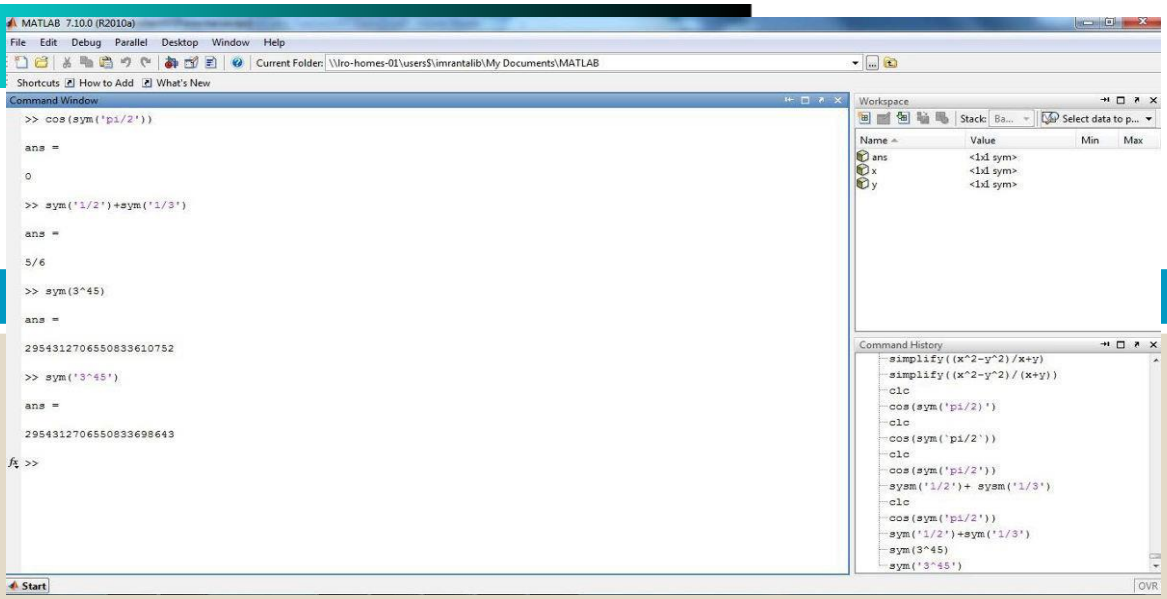


Figure: (6) The Matlab Desktop display with sym command

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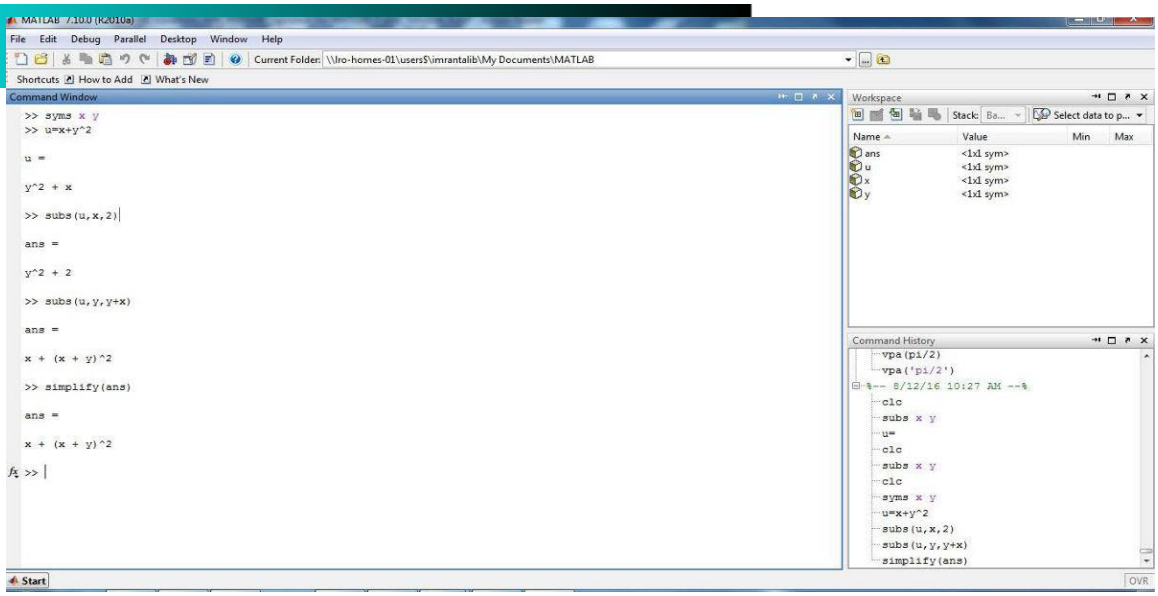


Figure: (7) The Matlab Desktop display with substitution command

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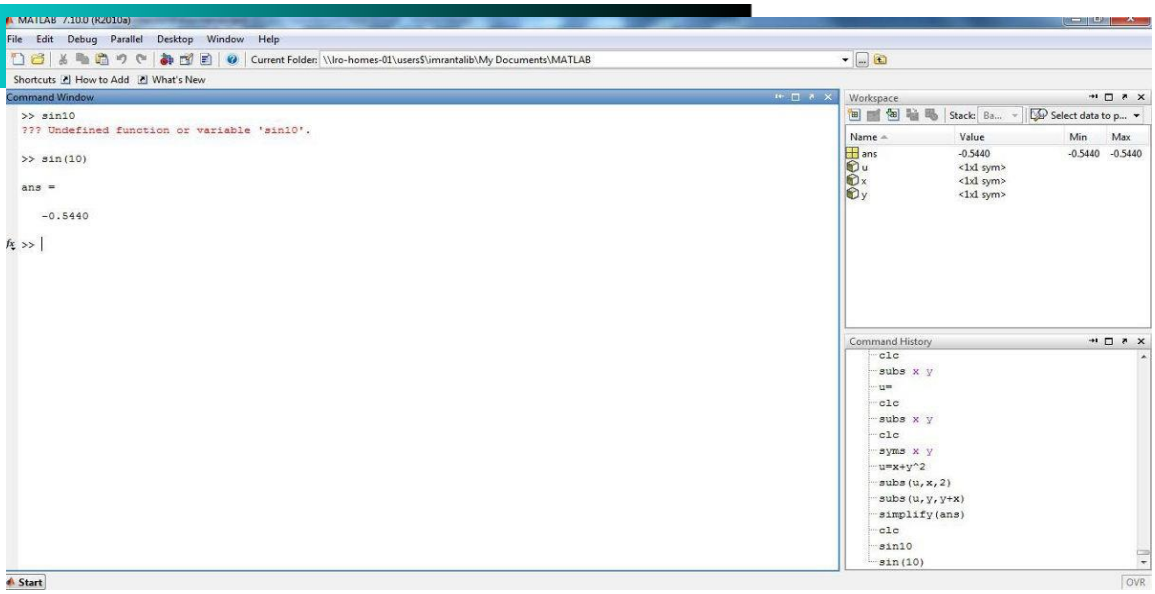


Figure: (8) The Matlab Desktop display while removing error

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Expand the following algebraic expressions using expand command:

- $(x + y)^{10}$
- $(x + y)^3 + (x + y)^2$

Simplify the following quantities:

- $2 + 5 - 4 + 10^{10} + 7^2$.
- $2/5 + 4/10 - 100^2$

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1 Doing Mathematics with MATLAB II
- Variables and Assignments
- Vectors
- Built-in Functions
- User Defined Functions
- Solving Algebraic Equations

2 Exercise

1 Doing Mathematics with MATLAB II

2 Exercise

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In MATLAB, = sign is used to assign values to a variable.

For example, to assign 7 value to a variable x, we proceed as

» x=7
x =
7

From now to own-ward, whenever MATLAB sees the letter x, it will substitute the value 7.

For example, if y has been defined as a symbolic variable, then

» x=7
x =
7
» syms y » $x^3 + 2 * x * y + 10 * x$
ans =
$14 * y + 413$

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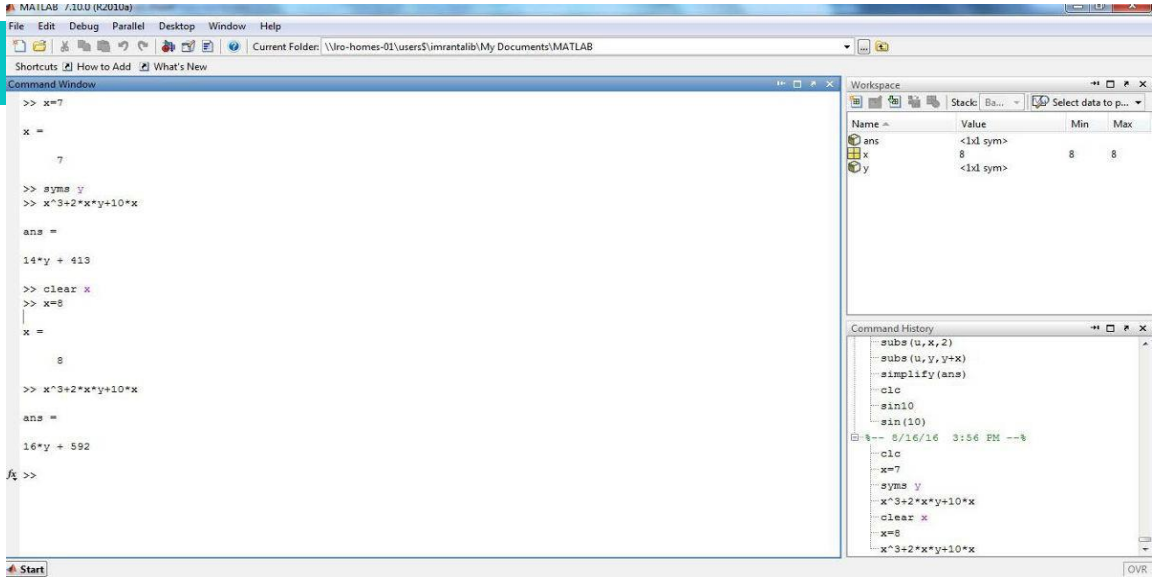


Figure: The Matlab Desktop display with Variable assign command

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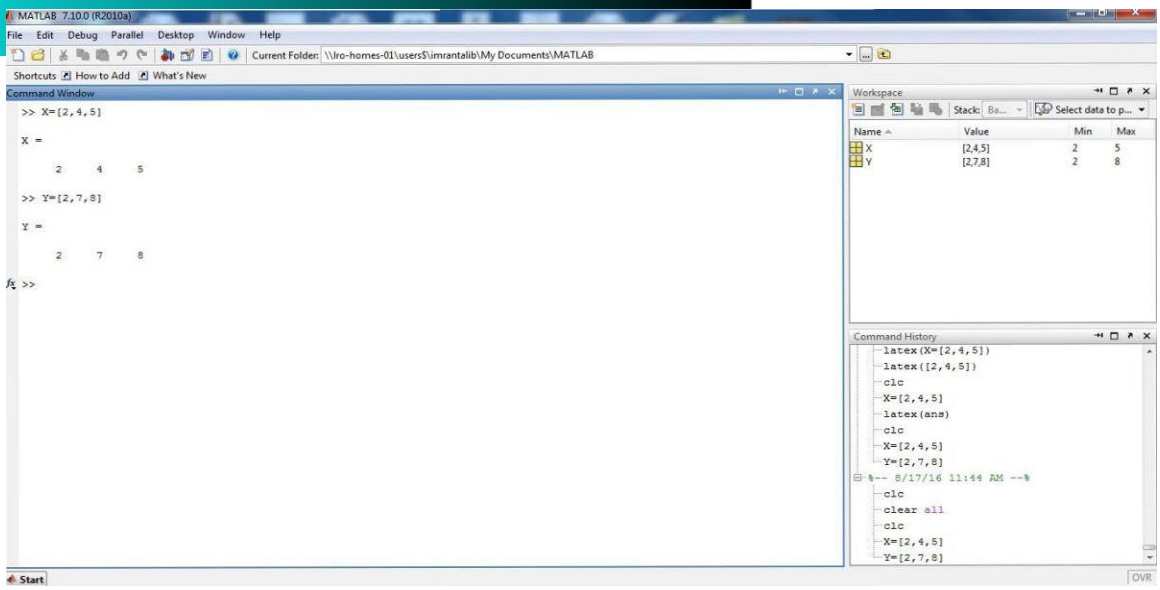


Figure: The Matlab Desktop display with Vector command

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To change a vector *Y* from a row vector to a column vector, simply put a prime (') after *Y*, as

```
» Y'
ans =
1
3
5
7
9
11
```

You can also extract the entries of vector *Y*. For example, the entry 5 can be extracted as

```
» Y(3)
ans =
5
```

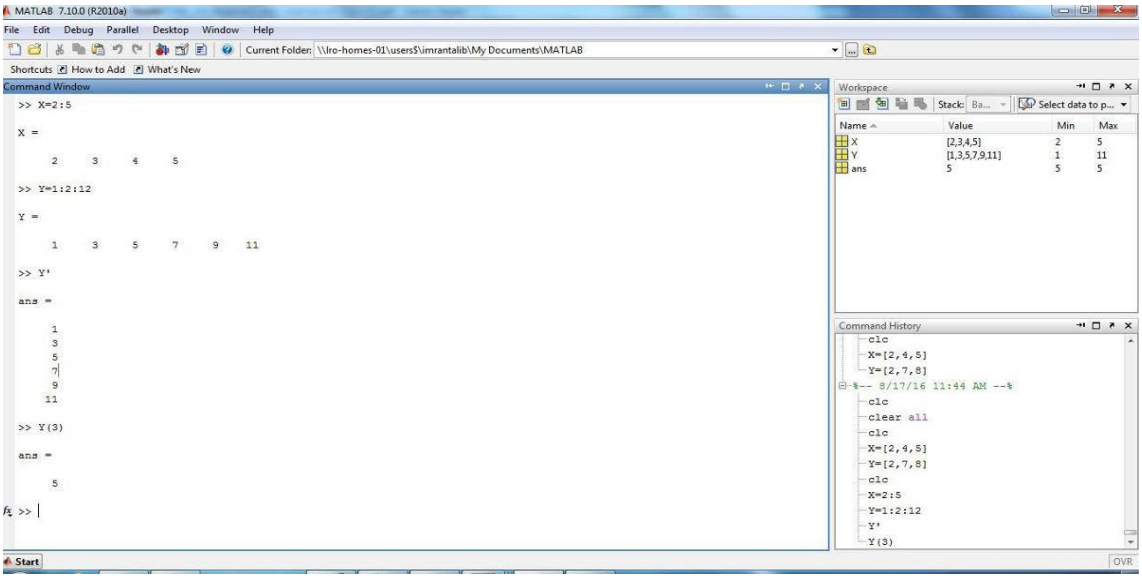


Figure:

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Mathematical operations can also be performed on vectors. For example, to square the elements of of the vector Y,use the following command

$Y.^2$

ans =

1 9 25 49 81 121

Similarly you can perform other mathematical operations, see the Figure 4

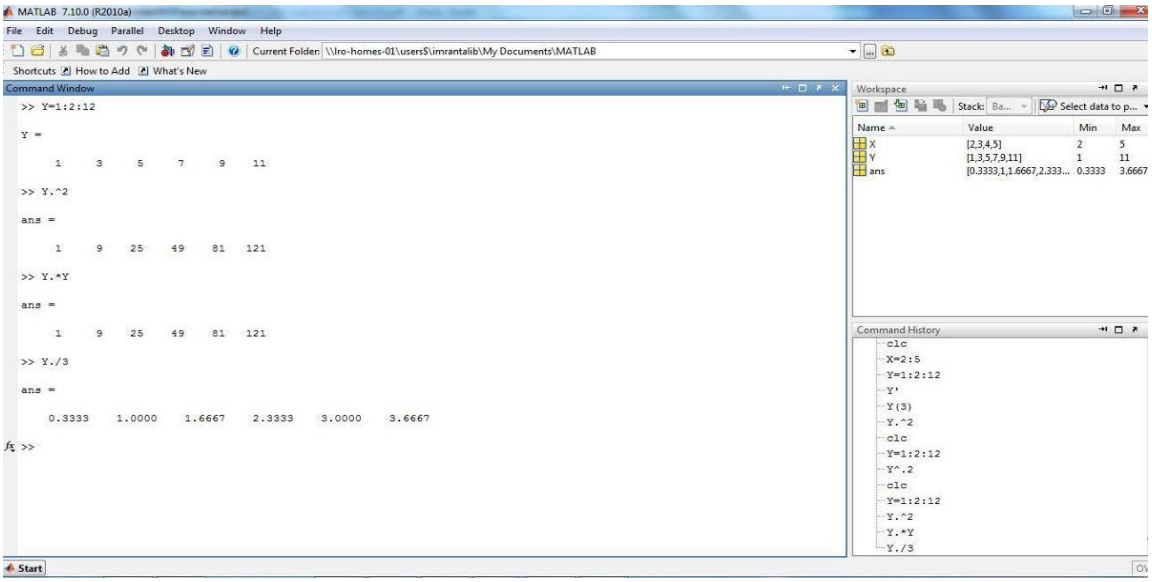


Figure:

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MATLAB has all the usual elementary functions built in. For example, exp(x) indicates the exponential function of x.

log(x) represents the natural logarithm function of the argument x.

sqrt is used to take the square root of an entity.

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Once the function is defined, it can be evaluated at any real number. For example

To calculate the value of x^3 at $x = 10$, we will proceed as

```
» f = @(x)x^3
f =
@(x)x^3
» f(10)
ans =
1000
```

See the Figure 5

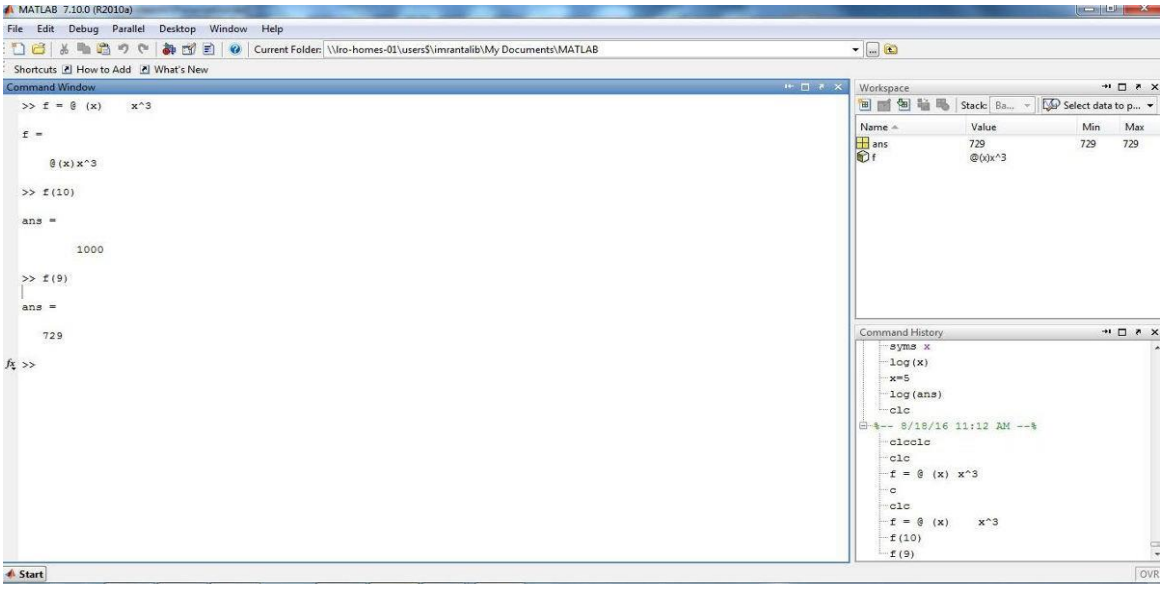


Figure:

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This command can also be used to define functions of several variables. For this we proceed as

» $h = @(x,y)x^2 + y^2$
h =
$@(x,y)x^2 + y^2$

$h(x,y) = x^2 + y^2$ can also be evaluated at different values of x and y as,

» h(1,2)
ans =
5

See the Figure 6

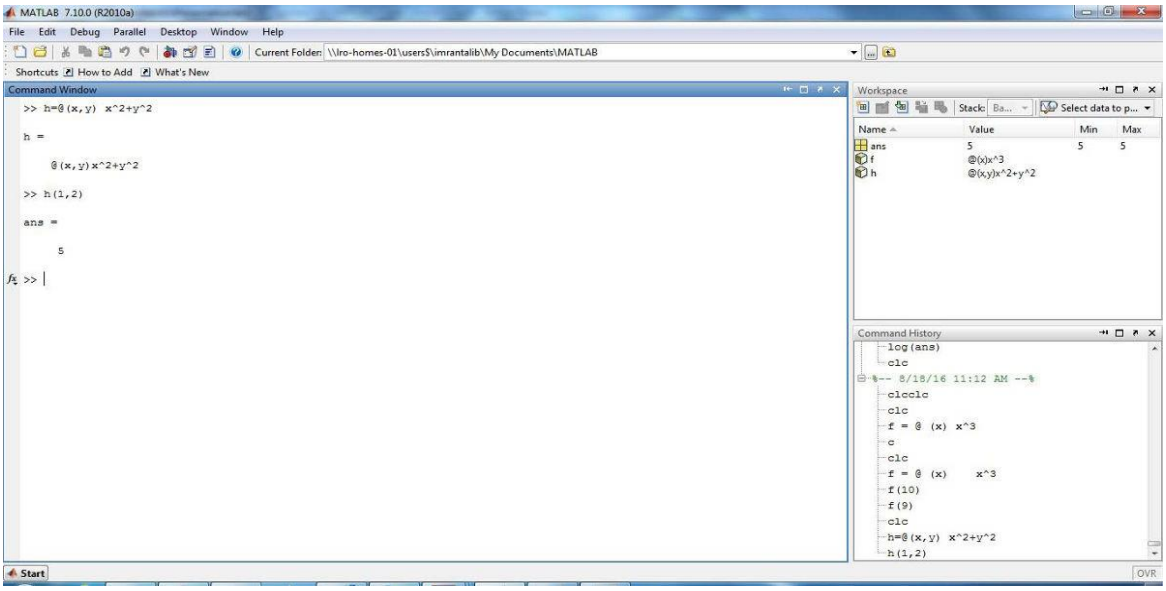


Figure:

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Now, we learn how to solve the Algebraic equations in MATLAB. The command solve or fzero is used for this purpose. For example,

The MATLAB solution of the equation $x^2 - 5x + 6 = 0$ can be find as

» $solve('x^2 - 5 * x + 6 = 0')$.

The MATLAB output is

ans =
2
3

Note that here the equation to be solved is specified as a string, that is, it is surrounded by a single quotes.

See the Command Window 7

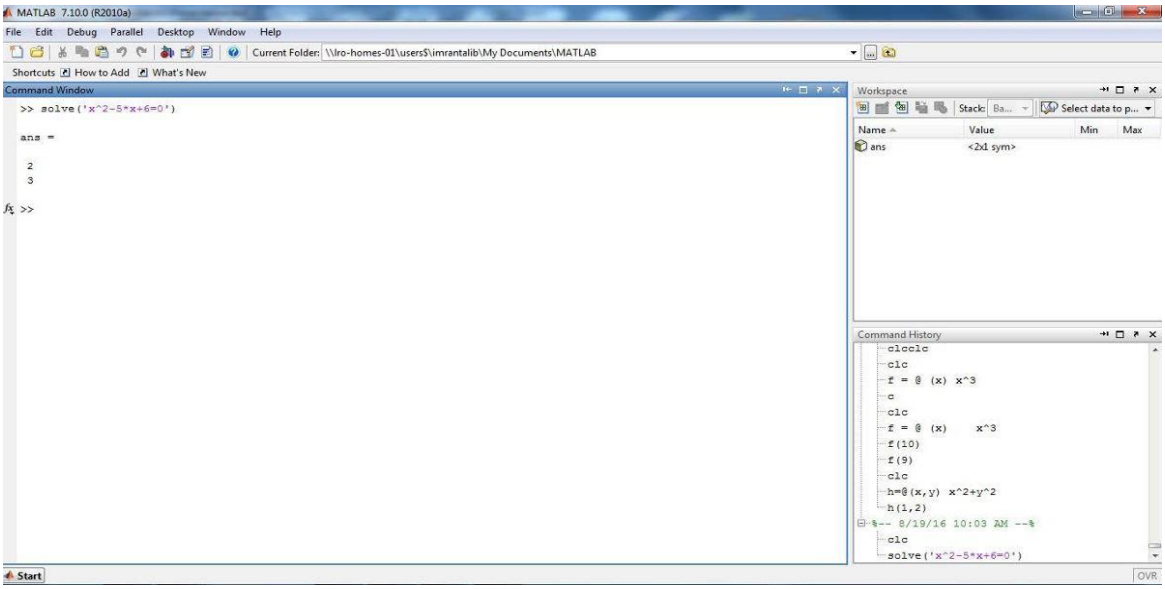


Figure:

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Note that, the input to solve can also be a symbolic expression. For instance

The MATLAB syntax for solving $x^2 - 3x = -7$ is:

syms x;

$solve(x^2 - 3 * x + 7)$

The MATLAB generated output is

ans =

$3/2 - (19^{(1/2)} * i)/2$

$3/2 + (19^{(1/2)} * i)/2$

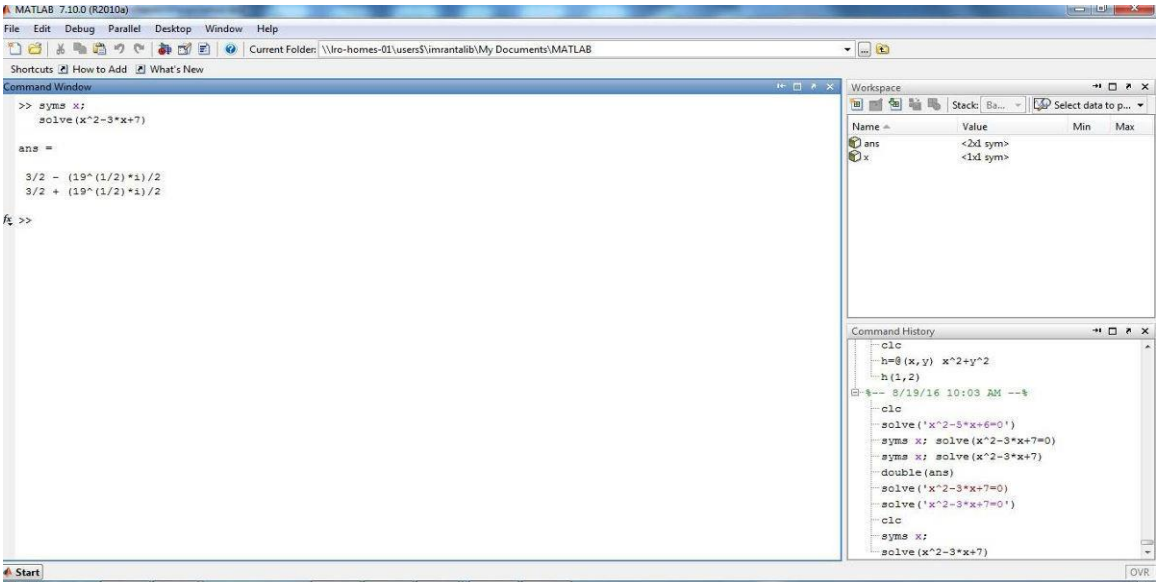


Figure:

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The solve command can also be used to solve higher degree polynomial equations, as well as many other types of equations.

It can also be used to solve equation having more than one variables. For example to find the solution of $x^2 - y = 2$ and $y - 2x = 5$, the following syntax is used:

» $[x, y] = solve('x^2 - y = 2','y - 2 * x = 5')$

The MATLAB generated output is:

x =

$2 * 2^{(1/2)} + 1$

$1 - 2 * 2^{(1/2)}$

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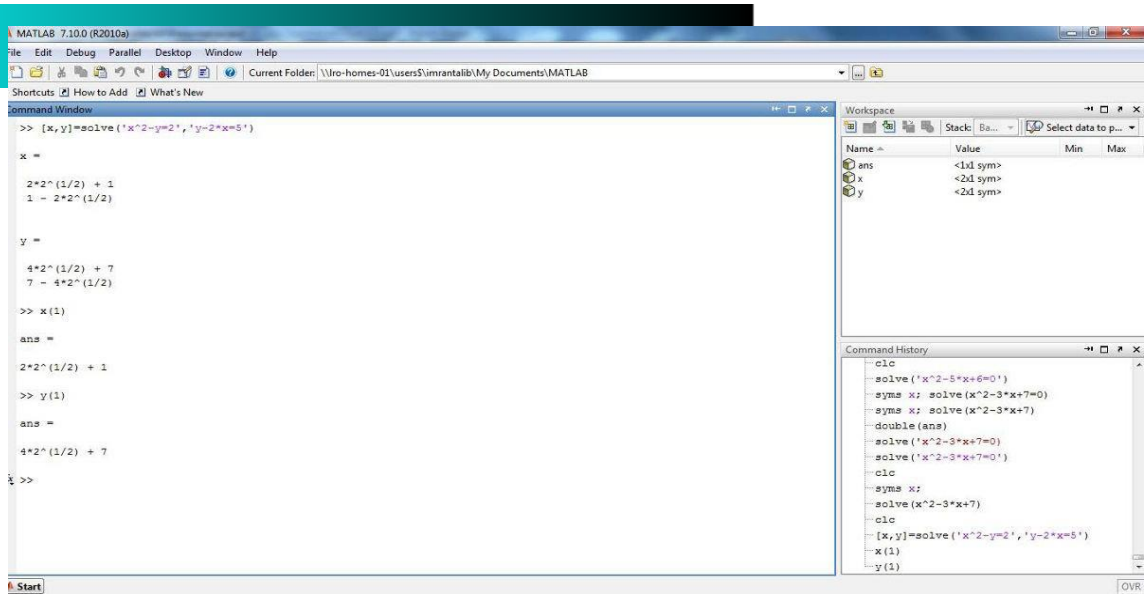


Figure: The Matlab Desktop display with the solution of the system of algebraic equations

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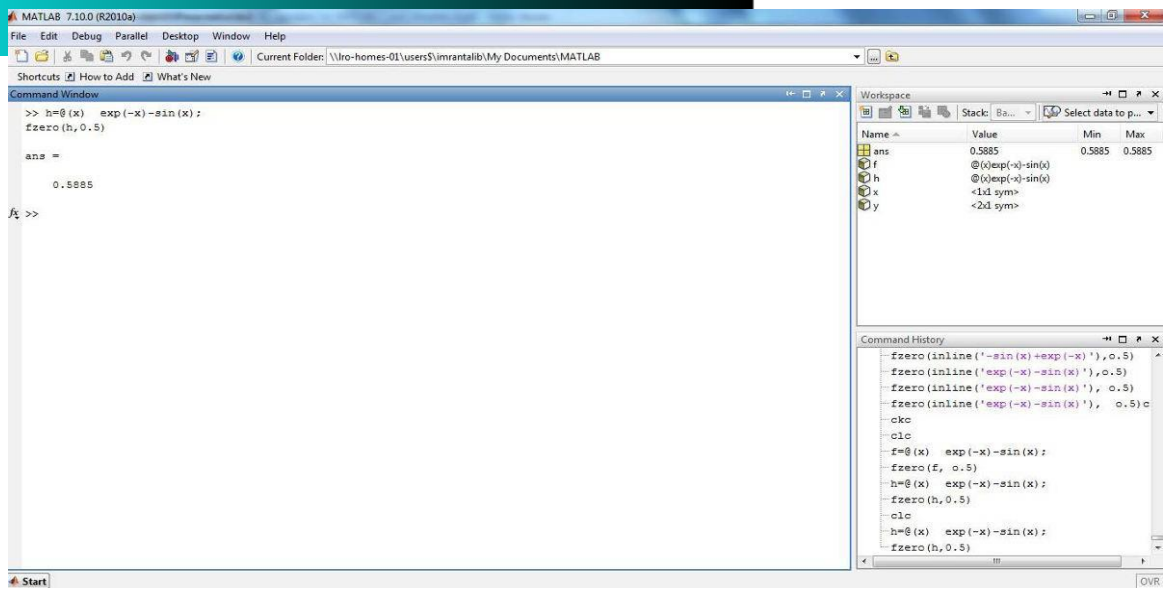


Figure: The Matlab Desktop display with fzero command

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Graphics Techniques

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1 Graphics Techniques in MATLAB

Lecture NO.04 2
ezplot Command
ezplot command with symbolic expression
ezplot command on anonymous function
Modification of Graphs
Example
Plot command
Graphs of Multiple Curves
Parametric Plots
Contour Plot
Exercise

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In this lecture, we learn the use of the basic plotting commands in MATLAB.

- ezplot command is used to plot the graph of a function of one variable.
- It expects a string, a symbolic expression or a function representing the function to be plotted.
- For example, the MATLAB command to draw the graph of the function x^3 on the interval $[-2, 2]$, using the string form of ezplot is:

  » ezplot('x^3',[-2,2]).

  Note that the result will display in a new window labeled "Figure1". Look at the Figure 1

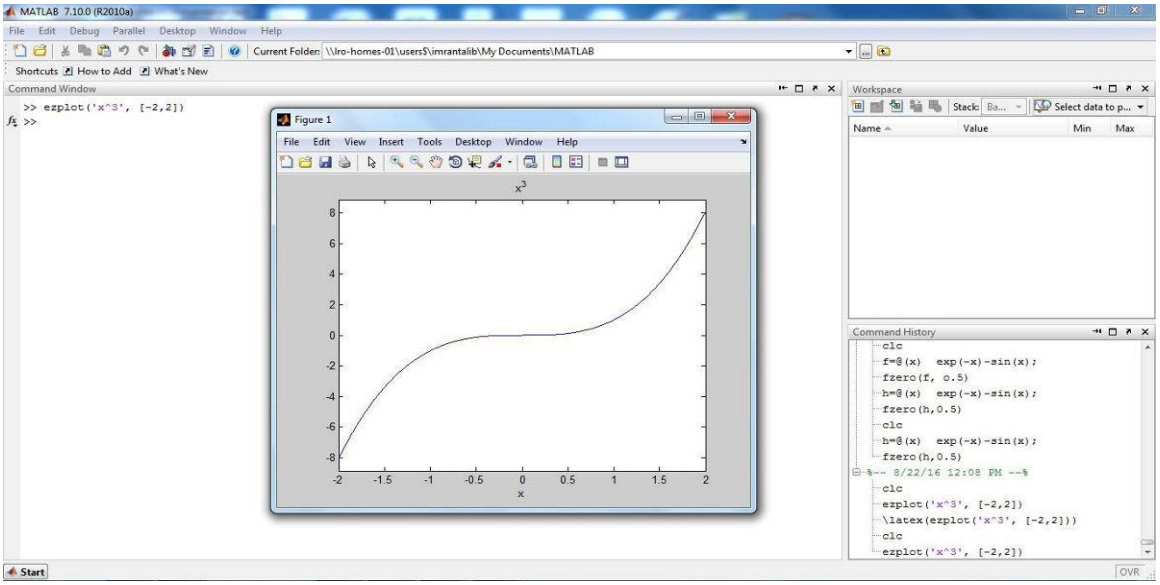


Figure:

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ezplot command can also accepts a symbolic expression. For example

The graph of x^3 can also be produced using the following input:

```
» syms x;
» ezplot(x^3, [-2,2])
```

See the Figure 2

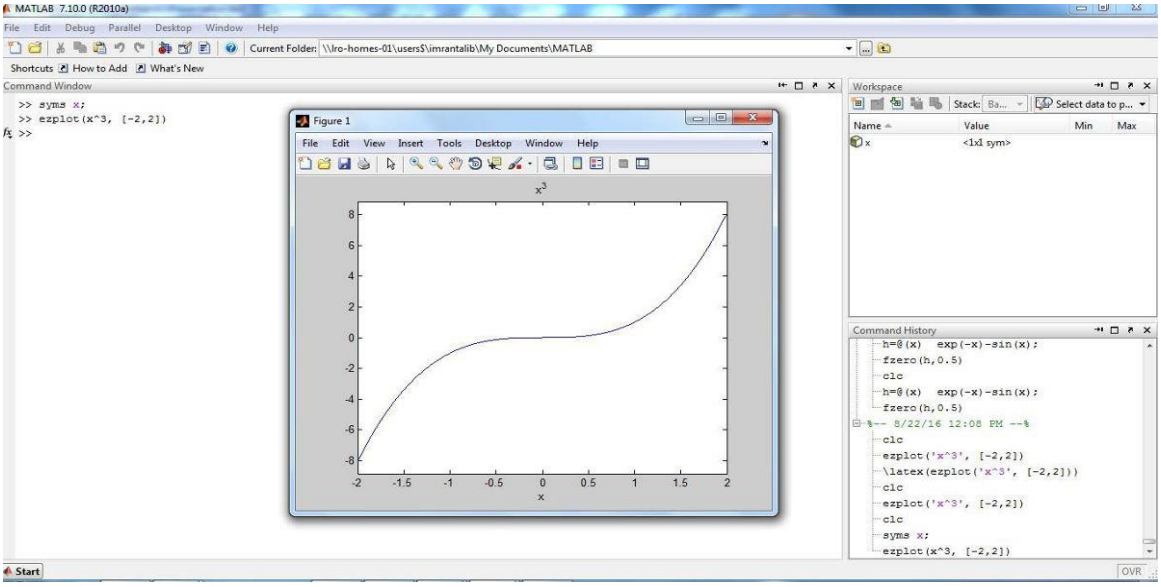


Figure:

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The graph of function x^3 can also be plotted by using an anonymous function as the argument of ezplot. For this case the input will be:

» ezplot(@(*x*)*x*.^3, [-2,2]).

See the Figure 3

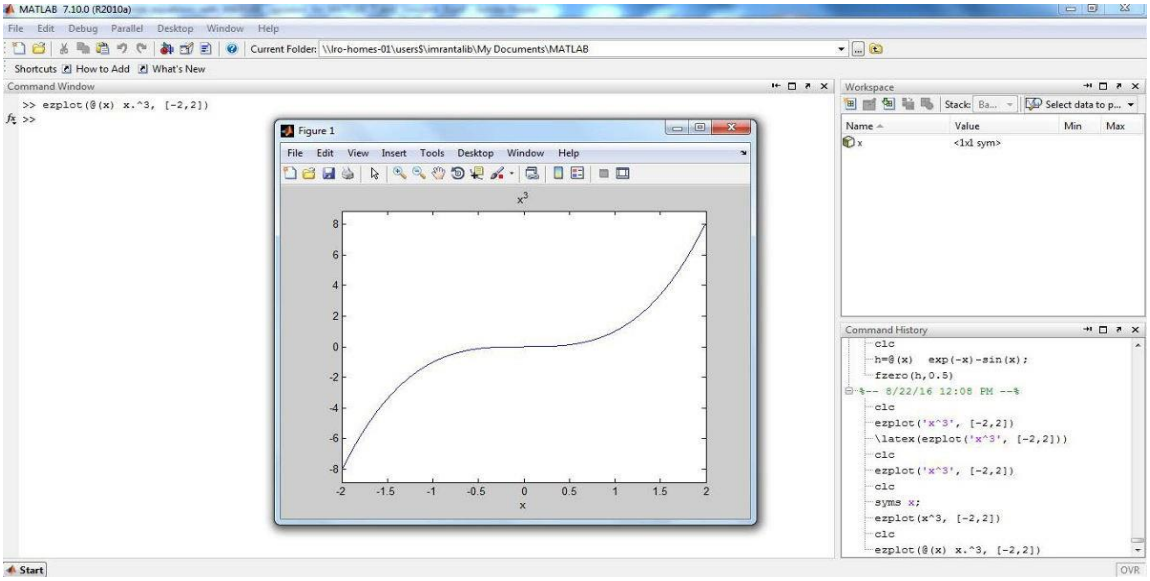


Figure:

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Graphs can be modified in a number of ways. For example

The title of the graph can be inserted.

Different types of labels can be given to both axis.

The region of the Graph can be seen in the form of grid.

To accomplish this task, simply type the following commands in the Command Window

```
syms xy;
y=ezplot(x^3, [-2,2])
xlabel('x');
ylabel('y');
title('The graph of y = x^3');
grid; ee the output 3
```

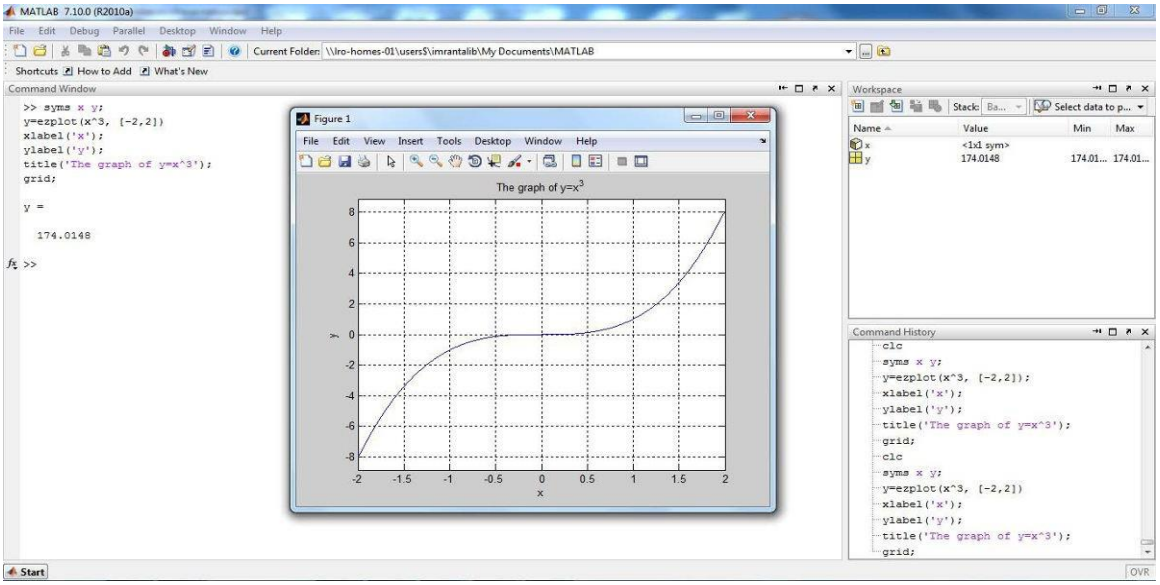


Figure:

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Example:

Plot the graph of the function $y = x^2$ on $[-3, 3]$ using ezplot command considering x and y to be symbolic variables.

Label the horizontal axis as "VU" and vertical axis as "IT".

Keep the title of the graph as "Parabola"

Solution:

The following syntax can be used to attain the required graph:

```
» syms x,  y;
y=ezplot(x^2, [-3,3])
xlabel('VU');
ylabel('IT');
title('Parabola');
grid;
```

Also see the Graph 5

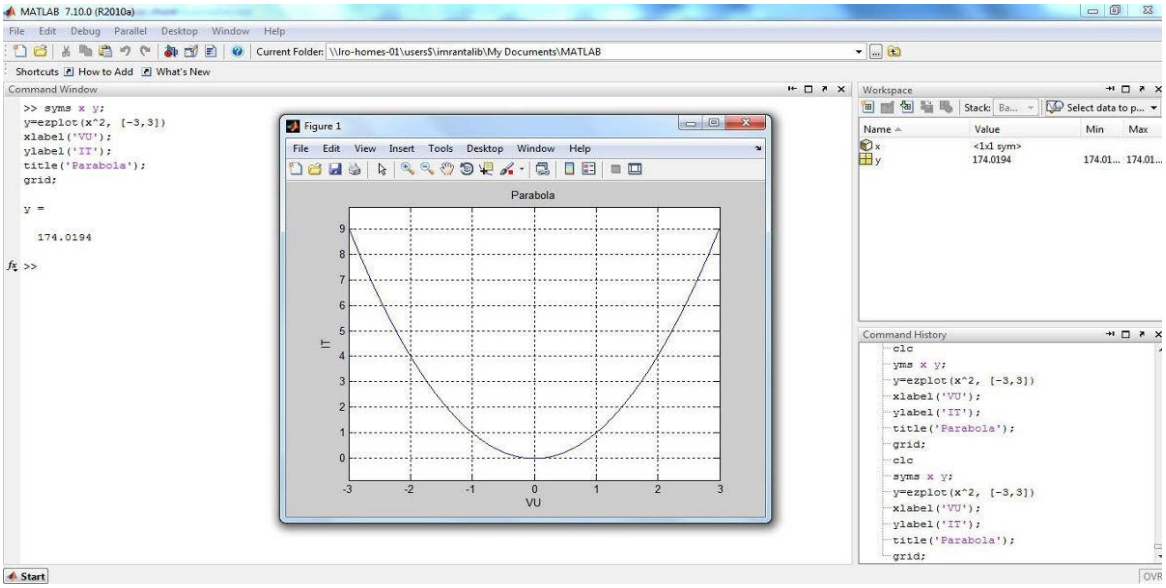


Figure:

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This command works on the vectors of numerical data.

The syntax is plot(X,Y).

X and *Y* are the vectors of same length.

This command considers the vectors *X* and *Y* to be lists of *x* and *y* coordinates of successive point on a graph, and joins the points with line

Under Plot command the vectors $X = [3\ 5\ 7]$ and $Y = [9\ 11\ 13]$ are connected by line segments as (3,9) to (5,11) to (7,13).

See their graphical view in Figure 7.

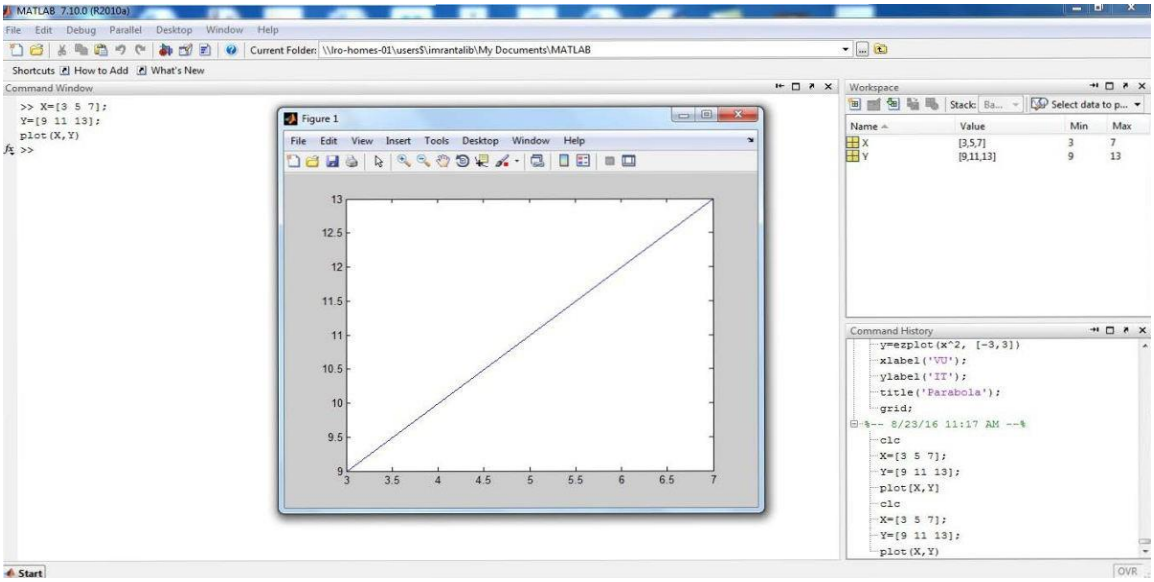


Figure:

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Example:

Using the plot command, plot the graph of $x^2 + x + 1$ on the interval $[-2, 2]$ by taking increment of 0.1 of the domain set.

Label the horizontal axis with x and vertical axis with y.

Finally, gives the title "Parabola" to the graph of $x^2 + x + 1$.

Solutioin:

Firstly, we make a list X of x values.

For this we need the enough values of x to get surety about the smoothness of the resulting graph. That is why we are going to subdivide the

Thus a recipe for graphing the parabola is:

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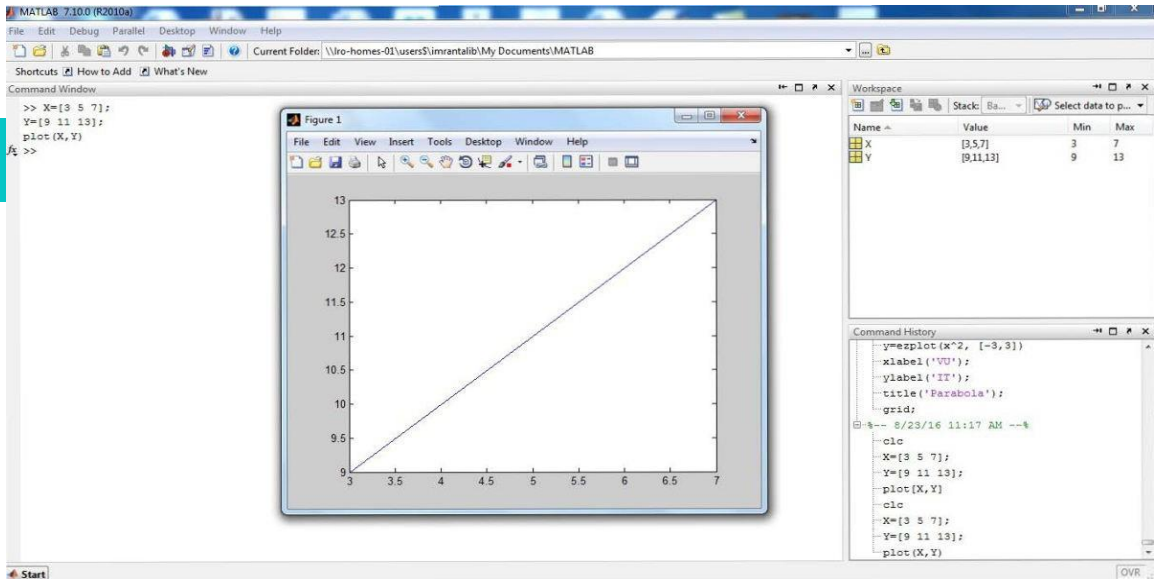


Figure: Plot command view of the Parabola

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Example:

Plot the graphs of e^{-x} and $sin(x)$ on the interval $[0, 8]$ using the ezplot, hold on, and hold off commands.

Label the horizontal axis with x and vertical axis with y.

Finally, gives the title "Multiple Graphs".

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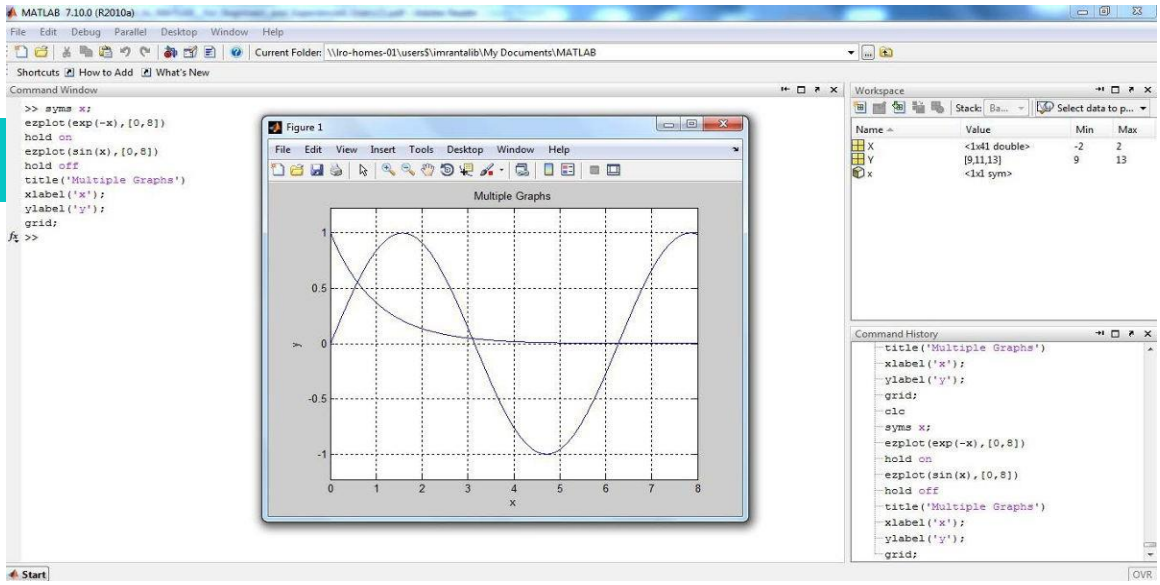


Figure: Ezplot command view of multiple curves

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Solution:

The recipe for graphing the multiple curves is:

```
» X=0:0.1:8;
plot(X , exp(−X ))
hold on;
plot(X , sin(X ))
hold off;
title('Multiple  Graphs')
xlabel('x');
ylabel('y');
grid;
```

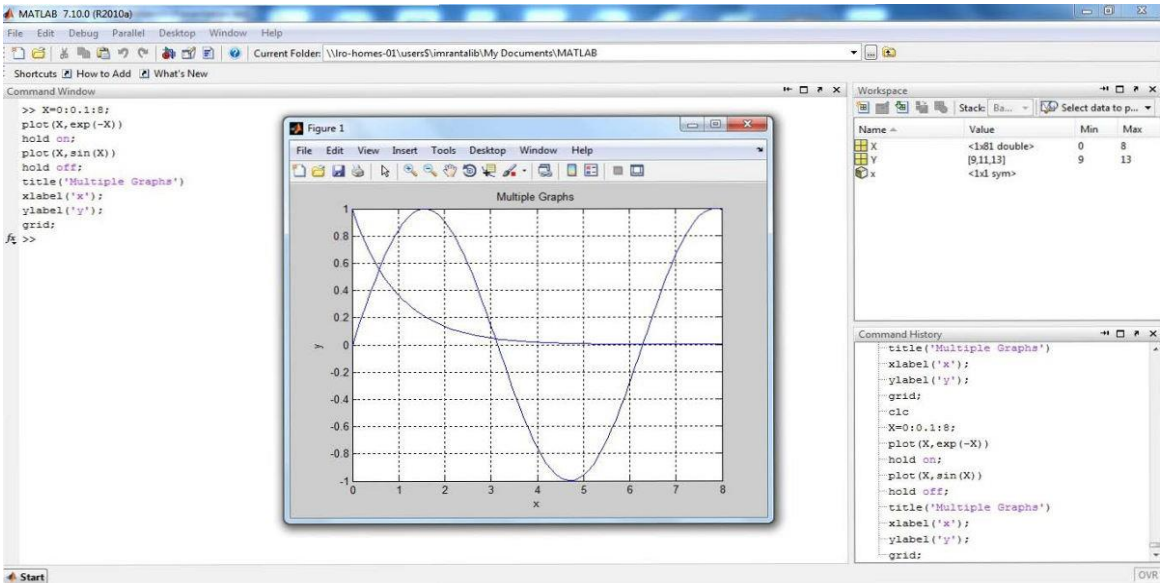


Figure:

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We observe the flexibility of the plot command by parametric plot of the circle centered at (0, 0), and radius 1.

The parametric representation of the circle can be expressed with the aid of equations $x = cos(2\pi t)$ and $y = sin(2\pi t)$, t varies from 0 to

The recipe for graphing the parametric plots is:

```
» t=0:0.1:1;
»  plot(cos(2*pi*t),sin(2*pi*t))
» xlabel('x');
» ylabel('y');
» title('Parametric plot of circle');
» grid;
» axis square;
```

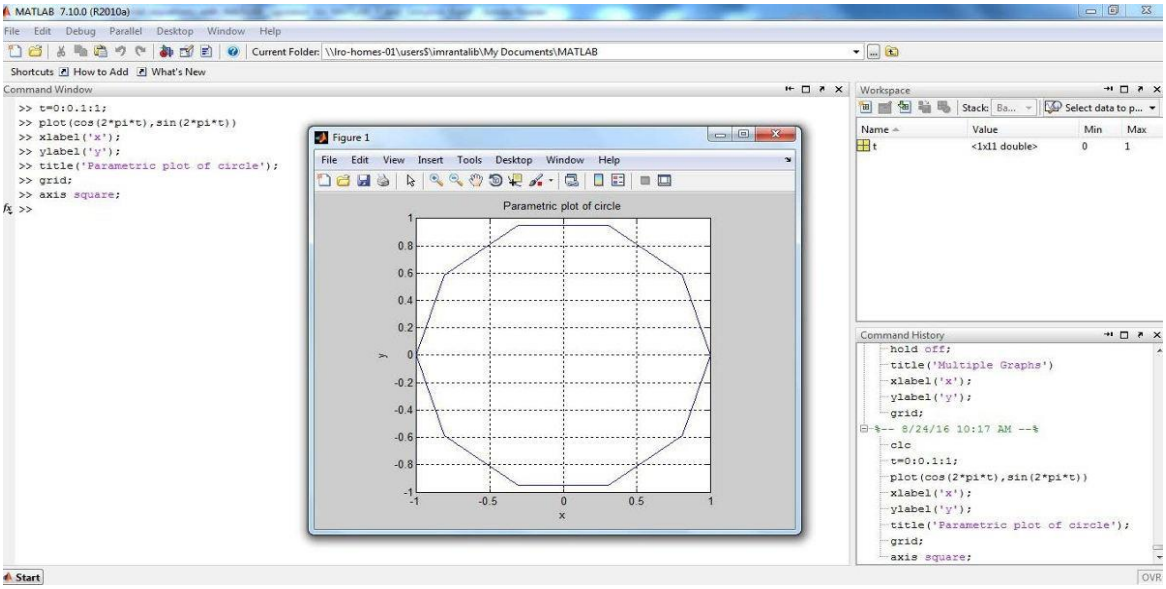


Figure:

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Example:

Plot the parametric graph of the circle with center at $(0, 0)$, radius 1, and with increment of 0.01 in the domain set.

Label the horizontal axis with x and vertical axis with y.

Make the shape of the graph square.

Keeps the title of the graph "Parametric plot of the circle".

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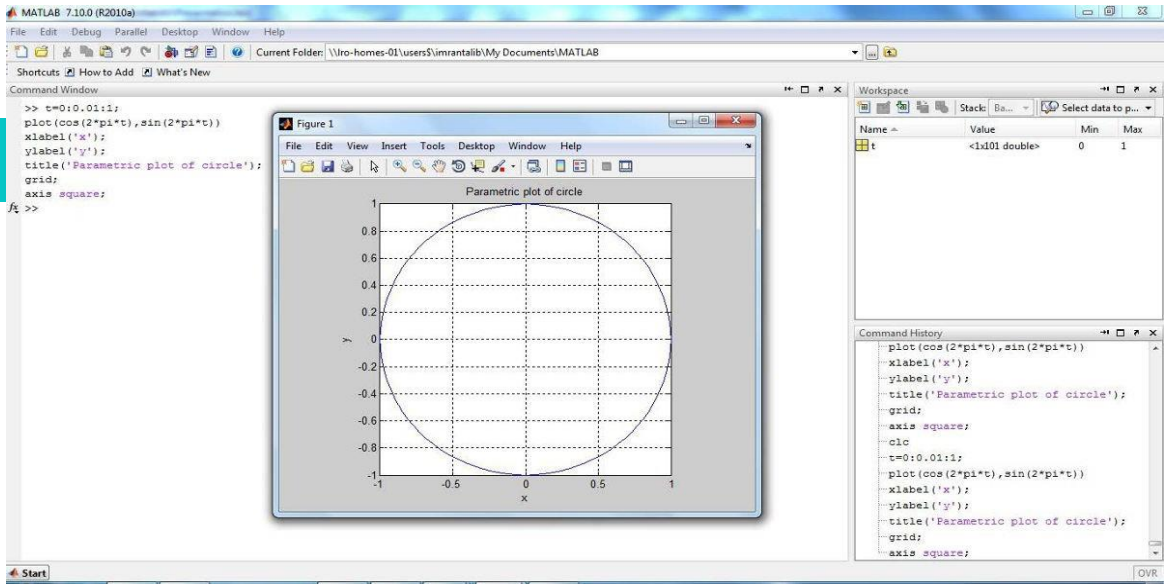


Figure: Parametric plot of the circle using plot command

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Example:

Plot the contour plots of the expression $x^2 + y^2$ using meshgrid and contour commands.

Label the horizontal axis with x and vertical axis with y.

Make the shape of the graph square.

Keeps the title of the graph "Contour plotting of squares".

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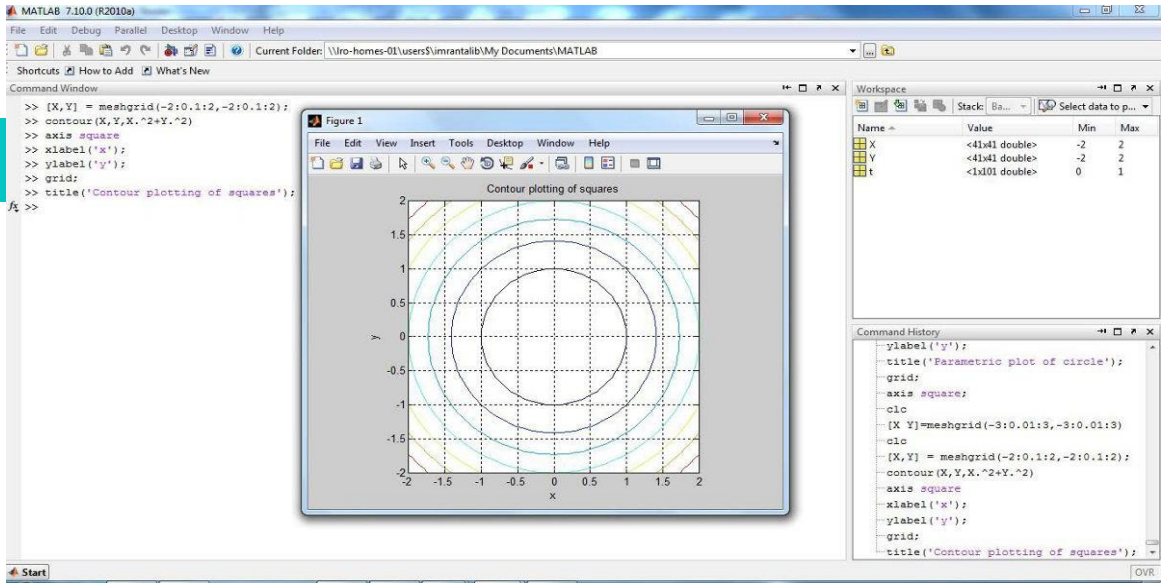


Figure: Contour plotting

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Plot the graph of the function $sin(x) + cos(x)$ on the interval [0, 1], using plot command.

Label the horizontal axis with x and vertical axis with y.
Make the shape of the graph square.
Keeps the title of the graph "The Graph of $sin(x) + cos(x)$ ".
Keeps the layout of the graph as grid.

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Plot the multiple curves of the functions $sin(x) + cos(x)$ and $e^{-x} + x^2$ on the interval $[0, 4]$, using plot command and without the use of hold on and hold off commands.

Label the horizontal axis with x and vertical axis with y.
Make the shape of the graph square.
Keeps the title of the graph "The Graph of Multiple curves using plot command ".
Keeps the layout of the graph as grid.

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Plotting in MATLAB

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Ezplot, Plot and FPlot

- We have used different plotting command toplot the curve in MATLAB.

- We will go through these command and check what are the differences between these.

ezplot Command

- ezplot(fun) plots the expression fun(x) over the default domain $-2\pi < x < 2\pi$, where fun(x) is an explicit function of only x.

- ezplot(fun)

- ezplot(fun,[xmin,xmax])
- ezplot(funx,funy)

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ezplot

For example

Passing the Function as a Character Vector or String

```
ezplot('x^2')
```

```
Ezplot('x^2-y^4')
```

Passing a Function Handle

```
fh = @(x,y) x.^2 + y.^3 - 2*y - 1;
ezplot(fh)
```

plot Command

- Plot function also plots a 2-D line plot.
- plot(X,Y) creates a 2-D line plot of the data in Y versus the corresponding values in X.

- plot(X,Y,LineSpec) sets the line style, marker symbol,

and color.

- plot(X1,Y1,...,Xn,Yn) plots multiple X, Y pairs using the same axes for all lines.

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plot

- plot(X1,Y1,LineSpec1,…,Xn,Yn,LineSpecn) sets the line style, marker type, and color for each line. You can mix X, Y, LineSpec triplets with X, Y pairs.
- For example, plot(X1,Y1,X2,Y2,LineSpec2,X3,Y3).

- x = 0:pi/100:2*pi;
- y = sin(x);
- plot(x,y)

- x = linspace(-2*pi,2*pi);
- y1 = sin(x);
- y2 = cos(x);

- plot(x,y1,x,y2)

- Y = magic(4)
- plot(Y)

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Specify Line Style

- x = 0:pi/100:2*pi;
- y1 = sin(x);
- y2 = sin(x-0.25);
- y3 = sin(x-0.5);

- figure
- plot(x,y1,x,y2,'--',x,y3,':')

Specify Line Style, Color, and Marker

- x = linspace(0,10);
- y = sin(x);
- plot(x,y,'-o','MarkerIndices',1:5:length(y))

Line Style

- Line Style Description
- - Solid line
- -- Dashed line
- : Dotted line
- -. Dash-dot line

Marker

- Marker Description
- 'o' Circle
- '+' Plus sign
- '*' Asterisk
- '.' Point
- 'x' Cross
- '_' Horizontal line
- '|' Vertical line
- 's' Square
- 'd' Diamond

- '^' Upward-pointing triangle
- 'v' Downward-pointing triangle
- '>' Right-pointing triangle
- '<' Left-pointing triangle
- 'p' Pentagram
- 'h' Hexagram

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Color

Color	Description
y	Yellow
m	magenta
c	cyan
r	red
g	green
b	blue
w	white

fplot Command

- Plot expression or function
- fplot(f) plots the curve defined by the function y = f(x) over the default interval [-5 5] for x.

- fplot(f,xinterval) plots over the specified interval. Specify the interval as a two-element vector of the form [xmin xmax].

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- fplot(funx,funy) plots the curve defined by x = funx(t) and y = funy(t) over the default interval [-5 5] for t.
- fplot(funx,funy,tinterval) plots over the specified interval. Specify the interval as a two-element vector of the form [tmin tmax].
- fplot(___,LineSpec) specifies the line style, marker symbol, and line color. For example, '-r' plots a red line. Use this option after any of the input argument combinations in the previous syntaxes.

Examples

- Plot sin(x) over the default x interval [-5 5].
- fplot(@(x) sin(x))
- Plot Parametric Curve

- xt = @(t) cos(3*t);
- yt = @(t) sin(2*t);
- fplot(xt,yt)

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Specify Plotting Interval and Plot Piecewise Functions

```
fplot(@(x) sin(x+pi/5),'Linewidth',2);
hold on
fplot(@(x) sin(x-pi/5),'--or');
fplot(@(x) sin(x),'-.*c')
hold off
```

Plot Multiple Lines in Same Axes

- fplot(@(x) sin(x))
- hold on
- fplot(@(x) cos(x))
- hold off

3D Curve Plotting in MATLAB

fplot3

- 3-D parametric curve plotter.

- fplot3(funx,funy,funz)

- fplot3(funx,funy,funz,tinterval)
- fplot3(____,LineSpec)
- fplot3(____,Name,Value).

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Example

- Plot the 3-D parametric line

x=sin(t)

y=cos(t)

z=t

over the default parameter range [-5 5].

```
xt = @(t) sin(t);
yt = @(t) cos(t);
zt = @(t) t;
fplot3(xt,yt,zt)
```

Specify Parameter Range

x=e^(–t/10)sin(5t)

y=e(–t/10)cos(5t)

z=t

over the parameter range [-10 10] by specifying the fourth input argument of fplot3.

```
xt = @(t) exp(-t/10).*sin(5*t);
yt = @(t) exp(-t/10).*cos(5*t);
zt = @(t) t;
fplot3(xt,yt,zt,[-10 10])
```

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Specify Line Properties and Display Markers

- fplot3(@(t)sin(t), @(t)cos(t), @(t)t, [0 2*pi], 'LineWidth', 2)
- hold on
- fplot3(@(t)sin(t), @(t)cos(t), @(t)t, [2*pi 4*pi], '--or')
- fplot3(@(t)sin(t), @(t)cos(t), @(t)t, [4*pi 6*pi], '-.*c')
- hold off

plot

- plot(X1,Y1,LineSpec1,...,Xn,Yn,LineSpecn) sets the line style, marker type, and color for each line. You can mix X, Y, LineSpec triplets with X, Y pairs.
- For example, plot(X1,Y1,X2,Y2,LineSpec2,X3,Y3).

- x = 0:pi/100:2*pi;
- y = sin(x);
- plot(x,y)

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fsurf

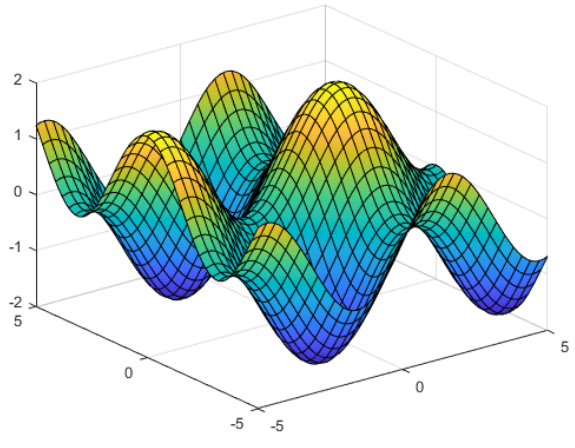
- fsurf(f)

  fsurf(f) creates a surface plot of the function z =f(x,y) over the default interval [-5 5] for x and y.
- fsurf(f,xyinterval)
- fsurf(funx,funy,funz)

  plots the parametric surface defined by x = funx(u,v), y = funy(u,v), z = funz(u,v) over the default interval [-5 5] for u and v.
- fsurf(funx,funy,funz,uvinterval)
- fsurf(___,LineSpec)
- fsurf(___,Name,Value)

- Plot the expression sin(x)+cos(y) over the default interval $-5<x<5$ and $-5<y<5$.

  fsurf(@(x,y) sin(x)+cos(y))



Specify Interval of Surface Plot and Plot Piecewise Expression

Plot the piecewise expression

erf(x)+cos(y) −5<x<0

sin(x)+cos(y) 0<x<5

over −5<y<5.

Specify the plotting interval as the second input argument of fsurf. When you plot multiple surfaces over different intervals in the same axes, the axis limits adjust to include all the data.

Code

- f1 = @(x,y) erf(x)+cos(y);
- fsurf(f1,[-5 0 -5 5])
- hold on
- f2 = @(x,y) sin(x)+cos(y);
- fsurf(f2,[0 5 -5 5])

- hold off

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Parameterized Surface Plot

Plot the parameterized surface

x=rcos(u)sin(v)
y=rsin(u)sin(v)
z=rcos(v)
wherer=2+sin(7u+5v)

for 0<u<2π and 0<v<π. Add light to the surface using camlight.

Code

```
r = @(u,v) 2 + sin(7.*u + 5.*v);
funx = @(u,v) r(u,v).*cos(u).*sin(v);
funy = @(u,v) r(u,v).*sin(u).*sin(v);
funz = @(u,v) r(u,v).*cos(v);
fsurf(funx,funy,funz,[0 2*pi 0 pi])
```

camlight

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fmesh

- fmesh(f)

  creates a mesh plot of the expression z = f(x,y) over the default interval [-5 5] for x and y.
- fmesh(f,xyinterval)
- fmesh(funx,funy,funz)

  plots the parametric mesh defined by x = funx(u,v), y = funy(u,v), z = funz(u,v) over the default interval [-5 5] for u and v.
- fmesh(funx,funy,funz,uvinterval)
- fmesh(___,LineSpec)

Example

- Plot a mesh of the input sin(x)+cos(y) over the default interval $-5<x<5$ and $-5<y<5$.

- fmesh(@(x,y) sin(x)+cos(y))

Parameterized Mesh Plot

Plot the parameterized mesh

x=rcos(s)sin(t)
y=rsin(s)sin(t)
z=rcos(t)
wherer=2+sin(7s+5t)

for 0<s<2π and 0<t<π. Make the mesh partially transparent using alpha.

Code

```
r = @(s,t) 2 + sin(7.*s + 5.*t);
x = @(s,t) r(s,t).*cos(s).*sin(t);
y = @(s,t) r(s,t).*sin(s).*sin(t);
z = @(s,t) r(s,t).*cos(t);
fmesh(x,y,z,[0 2*pi 0 pi]
```

M–Files
Exercise

M–Files
Exercise

Imran Talib VU

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1 M-Files
- Attributes of the M-Files
- Script M-files
- File saving procedure
- Execution or run of the Script file
- Display of the commands with results in Command Window
- Adding Comments
- Structuring Script M-File
- Function M-Files
- Loops

2 Exercise

1 M-Files

2 Exercise

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Up to now, we are working in Command Window. But for the complicated problems, Command Window is insufficient to produce the results.

To handle the coding of the complicated problems, a much better approach is to create an M-File.

MATLAB uses two different kinds of M-Files. One is script M-File and the other is function M-File.

We learn the use of both types of M-Files.

Imran Talib VU

M–Files
Exercise

Lecture NO.06 4

Attributes of the M–Files
Script M-files
File saving procedure
Execution or run of the Script file
Display of the commands with results in Command Window
Adding Comments
Structuring Script M–File
Function M–Files
Loops

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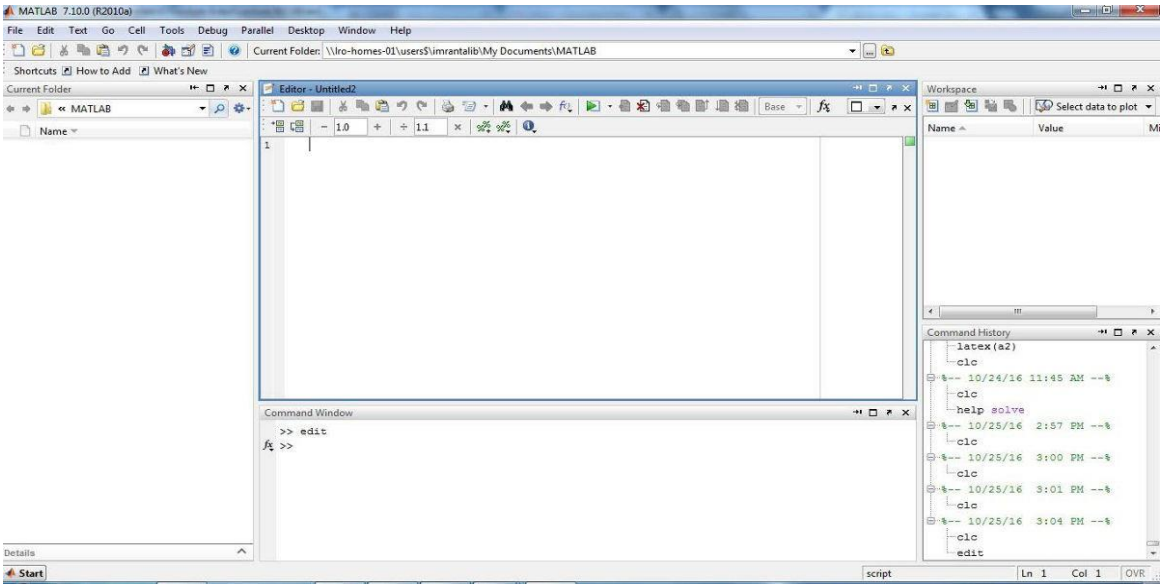


Figure: Display of the MATLAB desktop with Editor Window

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Solution:

Use Editor/Debugger to create the file containing the following lines:

$x = [0.1, 0.01, 0.001, 0.0001, 0.00001];$

$y = sin(x)./x.$

To see the output it is necessary to save and execute the above code.

The execution and saving procedure is explained in the next slide.

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The Script file can be executed or run by writing scriptfile1 in the Command Window.

Note that while the execution of the Script file, there is no need to write scriptfile1.m in the Command Window. Just write only scriptfile1.

The required out put will be displayed in the Command Window.

Remember that the Command Window does not give us liberty of editing.

But Editor Window has advantage of editing.

From now to onward, we will write our all codes in Editor Window instead of Command Window.

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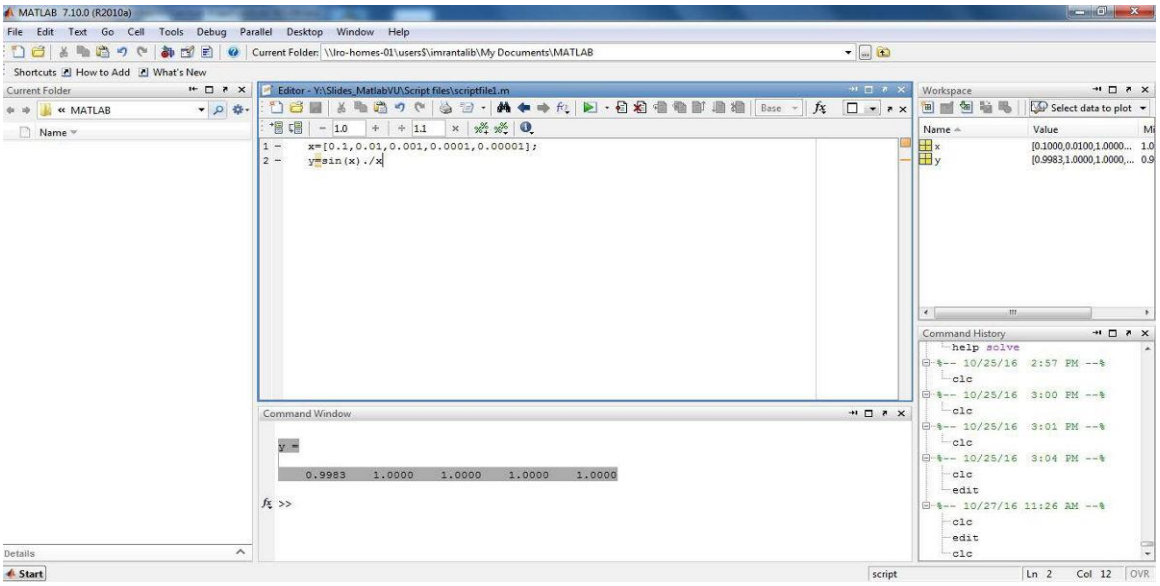


Figure: Computational values of $\frac{sin(x)}{x}$

Imran Talib VU

M–Files
Exercise

Lecture NO.06 12

Attributes of the M–Files
Script M-files
File saving procedure
Execution or run of the Script file
Display of the commands with results in Command Window
Adding Comments
Structuring Script M–File
Function M–Files
Loops

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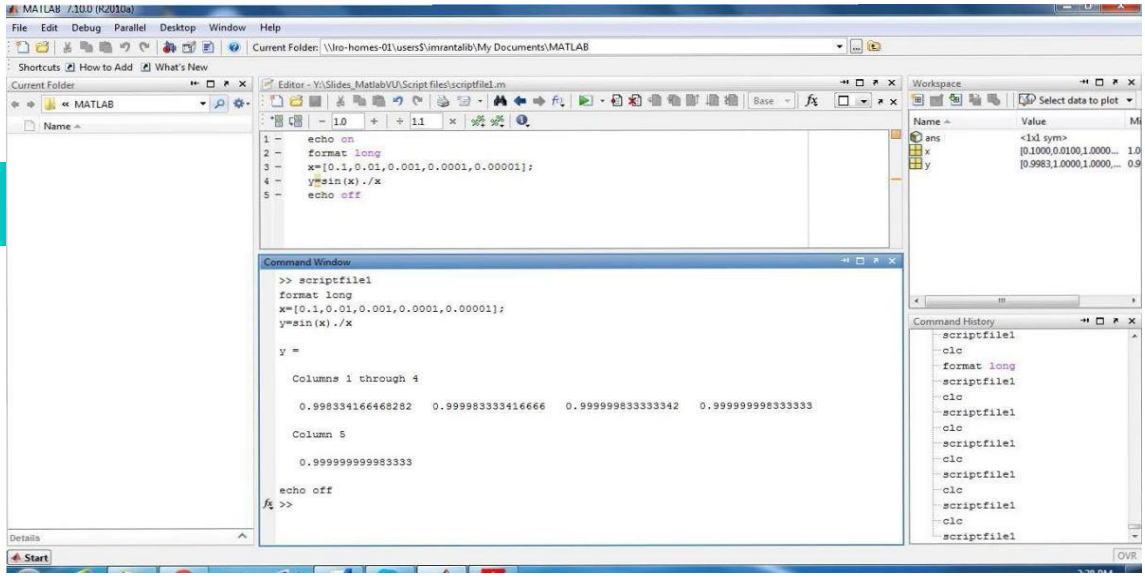


Figure: Display of the Commands along with the results in the Command Window using echo command.

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Solution with adding comments

echo on

format long

x=[0.1,0.01,0.001,0.0001,0.00001];

$y = sin(x)./x$

echo off

For more clarity see the next slide

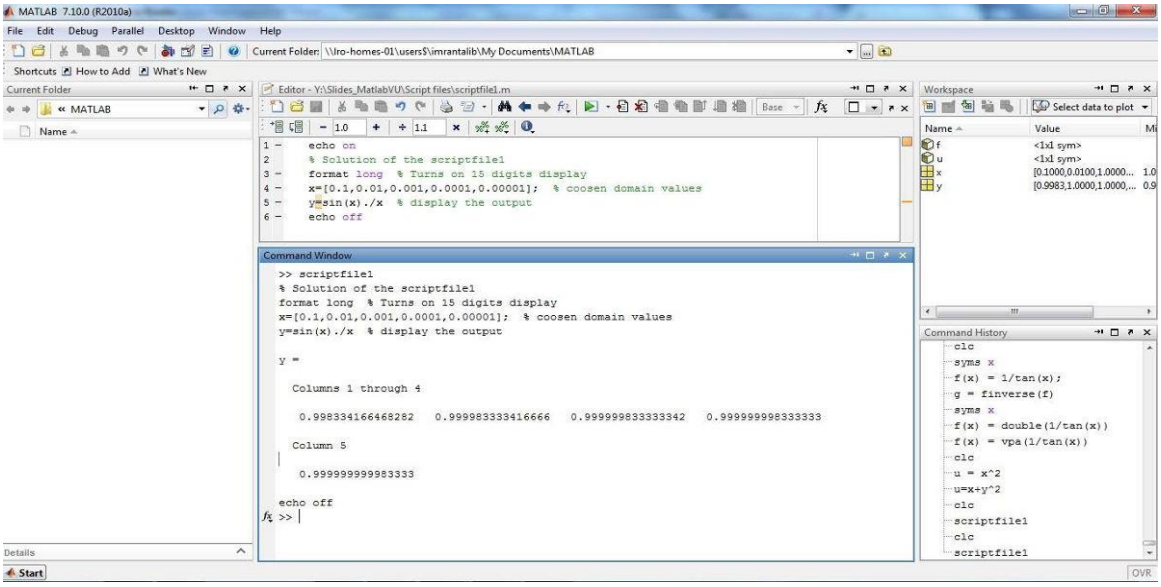


Figure:

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Make sure that your current MATLAB session is not affecting by other variables that had been defined in the last session. That is make sure that your present Script M-File is self contained, unaffected by other variables, and uncorrupted by left over graphics.

For this ensuring, clear all command should be used. This command ensure that previous definitions of variables do not affect the results.

Then in general your Script M-File should usually start with the clear all command.

You should also either include the close all command at the beginning of an M-File that creates graphics.

This command will automatically will close all graphics Windows and start with a clean slate.

For more clarity observe the new version of the above example with clear all and close all commands in the next slide.

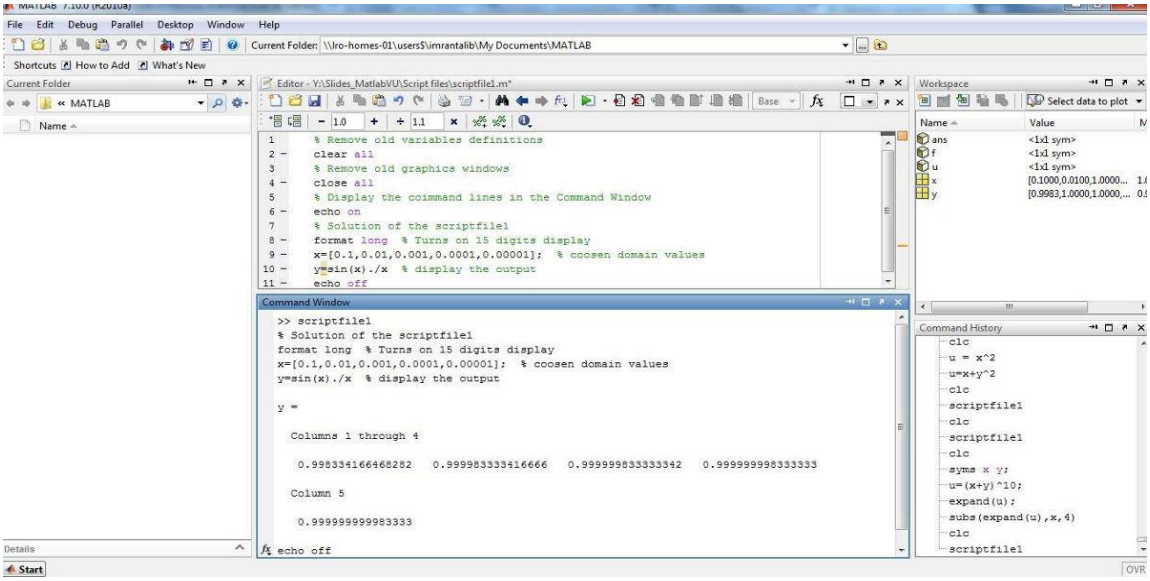


Figure:

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Sometimes, it is necessary to repeat a process several times for different input values of a parameter.

For instance, we can provide different inputs to a built-in function in order to find an output that meets a given criterion.

We have already defined our own functions using @ syntax or the inline command.

In many situations, however, it is more convenient to define a function using an M-File instead of an anonymous or inline function.

M-File allows us to extend the functionality of MATLAB by defining our own functions.

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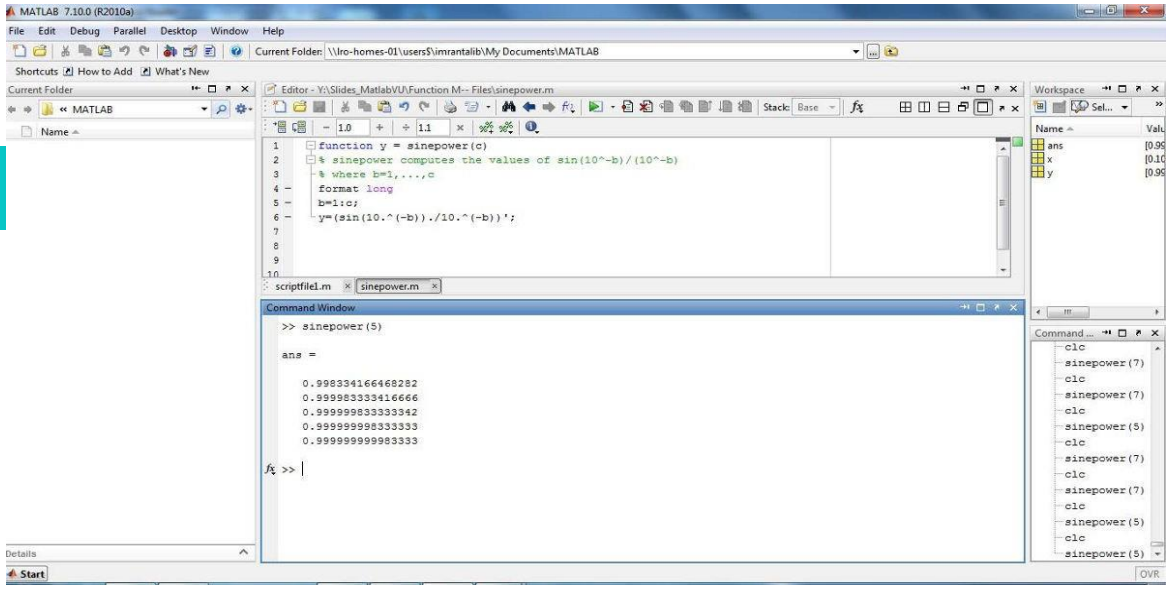


Figure: Solution of the previous example using Function M-File.

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Example:

Compute the values of $sin(x)$ at $x = 0, 1,$ and 2 using for loop.

Solution:

The recipe of the solution is as under

```
format long
for i= 0:2
y = sin(i)
end
y
```

For more clarity see the next slide.

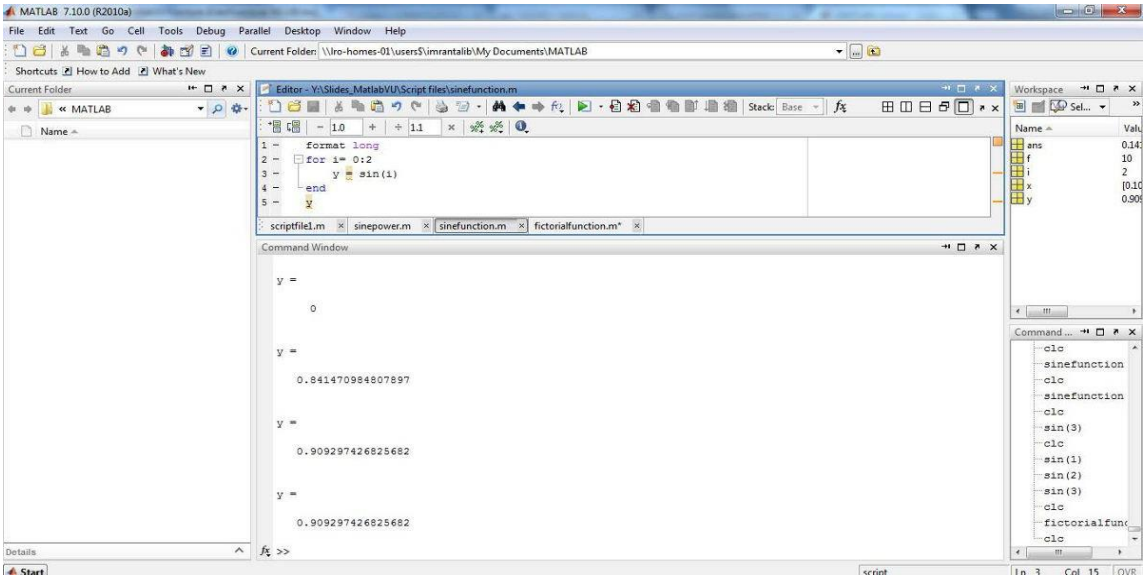


Figure:

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The loop begins with the for statement and ends with end statement. The command between those statements is executed a total of three times, once for each value of *i* from 0 to 2.

The Editor automatically colors the commands for and end in blue.

M–Files
Exercise

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Problem:

Find the first-order, second-order, and fourth-order derivatives of the function $f(x) = x^{10} + 2 * (x^2) + sin(x)$ using diff command.

Plot the multiple graphs of the outputs using ezplot command on the interval $[-\pi, \pi]$.

Label the horizontal axis with x and vertical axis with y.

Keeps the title "Multiple plot of the function and its derivatives"

All the coding should be written using Script M-File.

M–Files
Exercise

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Problem:

Find the first-order, fourth-order, and fifth-order derivatives of the function $f(x, y) = \ln x * y + sin(x^2)$ using diff command.

Plot the multiple graphs of the outputs using plot command on the interval $[0, 1]$ and without using hold on and hold off commands

Label the horizontal axis with x and vertical axis with y.

Keeps the title "Multiple plot of the function and its derivatives"

All the coding should be written using Script M-File.

Input Statements

Entering input into a program

- There are multiple ways to enter input variablesinto a MATLAB program. Three most common methods are as follows:

  I. Use hard-coded variables in MATLAB program.
  II. Use input statements within the program to prompt the

user in the MATLAB command window to enter input variable.

III. Use Menu statements within the program to prompt user to click on a button to select an input.

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Using Hard Coded Variables

- Whenever, we need a value in the program, we simply create a variable
- >>x=7;
- >>food='Pizza'

- Why not use hard coded variable all the time?
  I. Does not allow the user to choose different values.
  II. Program is only useful for one specific case.
  III. You might not know the needed value beforehand

Using Input Statements for Number

- To allow the user to choose a value is to use an input statement:

  >>variable=input('message')

Using Input Statements for String

- If you want to allow the user to enter a letter or a word instead of a number then the statement

>>variable= input('please enter a letter: ')

Please enter a letter: a

Error using input

Undefined function or variable a

- Add the second argument to the input statement to specify that the value will be string:

>>variable=input('Please enter a letter:', 's')

Examples

>>radius=input('Please enter the radius: ')

Please enter the radius: 3

>>disp(radius);

>>name=input('Please enter the name: ', 's')
Please enter the name: XYZ

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Using Menu Statement

- A menu statement can be used to allow the user to select form a set of predefined choices.
- When you use a menu statement, MATLAB will create a new window with a set of buttons.

```
>>variable1=menu('Message','opt1','opt2','opt3',...)
```

- The order you list the option in your message statement determines the order the buttons are displayed.
- The first button has value 1 and second 2 and so on.

Output Statements

Displaying the Outputs of a Program

There are a variety of ways to display the output of a MATLAB program. Three methods will be introduced initially:

1. Do not use ***suppression*** for a given statement in the MATLAB program.
2. Use ***display statements*** (disp) within the program that will display the desired outputs in the MATLAB command window.
3. Use ***fprintf statements*** within the program that will that will display the desired outputs in the MATLAB command window and/or write the resulting outputs to a file.

Unsuppressing Statement

- The simples way to display a value in MATLAB is not to suppress it:
- When you place a ; at the end of a line in MATLAB, you are telling MATLAB that you do not want to display the result in command window.
- If you leave ; at the end of a statement then MATLAB

will show you the result.

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Using Displaying Statement

- If you want to display additional information then you can use disp('message') command.
- You can use constant or variable in disp command to be displayed.

Example

If we want to display

```
>>radius=input('Please enter the radius: ');
>>Circumference= 2*pi*radius;
>>Area=pi*radius^2;
```

```
>>disp('Circumference : '); disp(circumferenece);
>>disp('Area'); disp(Area)
```

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fprintf statement

- If we want additional formatting, we can use the fprintf statement:

- Mix string with values
- Specify the number of decimal places
- **Insertion formats**
- %s –insert text
- %f-insert a floting number

Insertion formats

- %f

  Fixed-point notation (Use a precision operator to specify the number of digits after the decimal point.)

- %e

  Exponential notation, such as 3.141593e+00 (Use a precision operator to specify the number of digits after the decimal point.)

- %E

  Same as %e, but uppercase, such as 3.141593E+00 (Use a precision operator to specify the number of digits after the decimal point.)

- %g

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The more compact of %e or %f, with no trailing zeros (Use a precision operator to specify the number of significant digits.)

- %G

  The more compact of %E or %f, with no trailing zeros (Use a precision operator to specify the number of significant digits.)

Special Characters

Special Characters	Representation
Single quotation mark	``
Percent character	%%
Backslash	\\
Backspace	\b
New line	\n
Horizontal tab	\t
Vertical tab	\v

Example

```
>>day=`Wednesday`; weather=`Sunny`; temp=75;

>>fprintf(`Today is %s \n`, day)

>>fprintf(`The weather today is %s and the temperatue
```

is %i \n'`,weather,temp)

>>fprintf(`The weather today is %s and the temperatue is *%0.3f* \n'`,weather,temp)

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Print Double-Precision Values as Integers

- >>a = [1.02 3.04 5.06];
- >>fprintf('%d\n',round(a));

%d in the formatSpec input prints each value in the vector, round(a), as a signed integer. \n is a control character that starts a new line.

Print Literal Text and Array Values

>>A1 = [9.9, 9900];

>>A2 = [8.8, 7.7];

>>formatSpec = 'X is %4.2f meters or %8.3f mm\n';

>>fprintf(formatSpec,A1,A2)

Write Tabular Data to Text File

```
>>x = 0:.1:1;
>>A = [x; exp(x)];

>>fileID = fopen('exptest.txt','w');
>>fprintf(fileID,'%6s %12s\n','x','exp(x)');
>>fprintf(fileID,'%6.2f %12.8f\n',A);
>>fclose(fileID);
```


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1 Matrices in MATLAB

2 Matrices
Some useful operations on matrices

3 Matrix Algebra
Matrices Multiplication
Determinant of a matrix
Inverse of a matrix

4 Exercise

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1 Matrices in MATLAB
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1 Matrices in MATLAB
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Generating matrices as a list of its elements. The following basis conventions have to be taken into account:

The elements of a row are separated by commas or spaces.

A semi-colon indicates the end of each row.

The entire list of elements has to be surrounded by brackets [ ].

J

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Row vector can also be generated using Colon operator(;)

The entries in the generated Row vector iterates from the starting to the ending value with a default step of one.

In this case the square brackets are not necessary.

For example, a row vector having 2, 3, 4, 5, 6 entries can be generated as

» colvec=2:6

The necessary output is

colvec =
2 3 4 5 6

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Row vector is generated in different ways

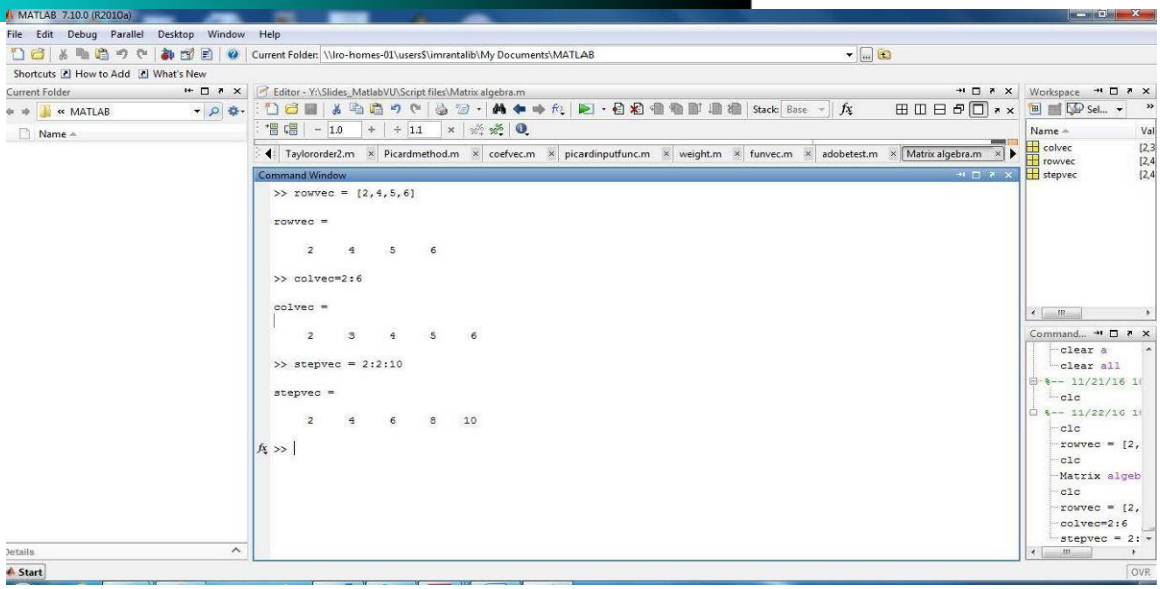


Figure: Display of the Row vector in different ways

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For example, the column vector with entries 2, 5, 10 can be generated as

» colvec=[2;5;10]

The required output is

colvec =

2

5

10

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Column vector is generated in different ways

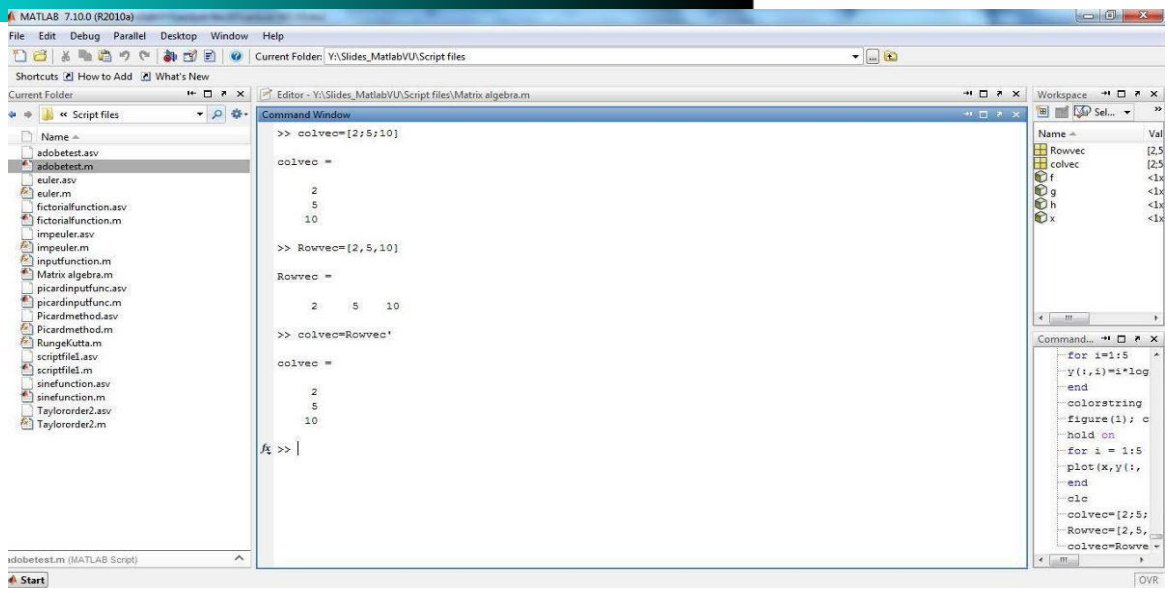


Figure: Display of the Column vector in different ways

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For example, the matrix

$$A = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & 0 \\ \frac{5}{6} & 0 & \frac{1}{6} \\ 0 & \frac{5}{6} & \frac{1}{6} \end{bmatrix}$$

in MATLAB is generated as

```
» A=[5/6,1/6,0;5/6,0,1/6;0,5/6,1/6]
A =
0.8333 0.1667 0
0.8333 0 0.1667
0 0.8333 0.1667
```

or

– –

– –

– –

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Generation of the matrix in MATLAB

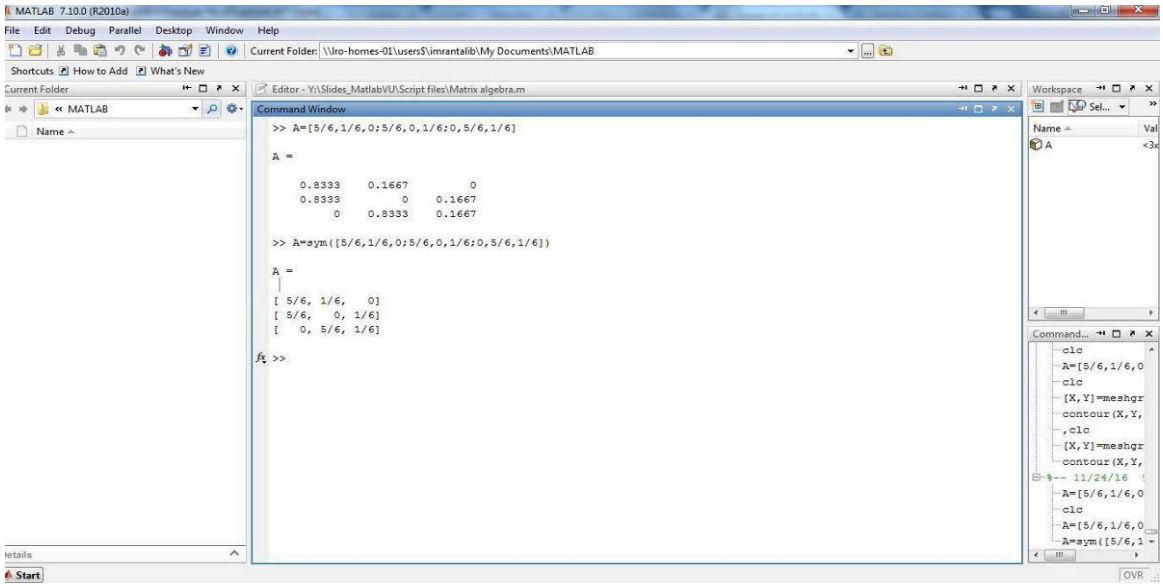


Figure: Graphical view of the matrices

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To compute the sum of Row vectors of a matrix, we should take the transpose, then compute the sum and finally transpose the result.

For example, the sum of the Row vectors of the matrix A is calculated in the next slide. Please see it.

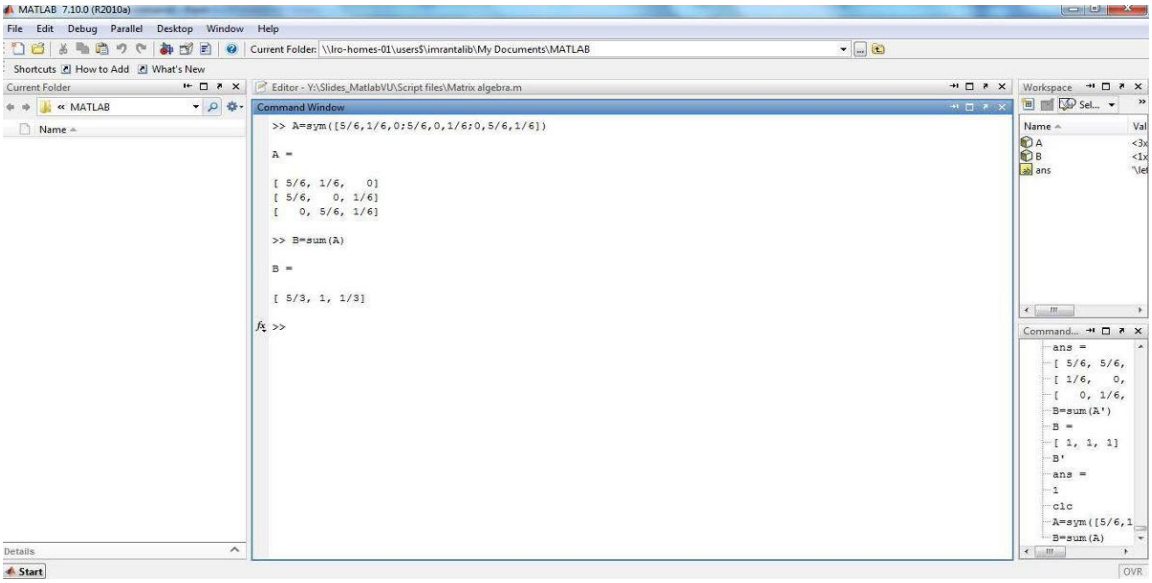


Figure:

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» A =

5/6	1/6	0
5/6	0	1/6
0	5/6	1/6

» A’=

5/6	5/6	0
1/6	0	5/6
0	1/6	1/6

» B=sum(A’)

B =

[1, 1, 1]

» B’

ans =

1

1

1.

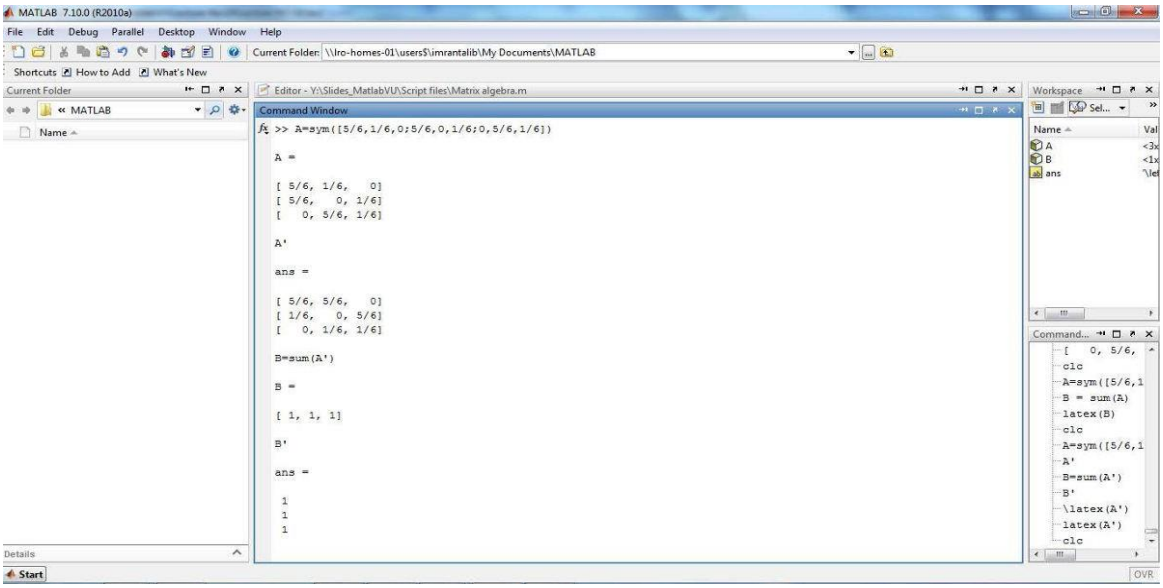


Figure:

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The diag command produces a column vector containing the elements of the main diagonal of a matrix.

For example,

```
» A =
    5/6   1/6   0
    5/6   0     1/6
    0     5/6   1/6
```

The MATLAB displays the following result:

```
» B=diag(A)
B =
5/6
0
1/6
```

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Sum of the entries of the last and the second column of A

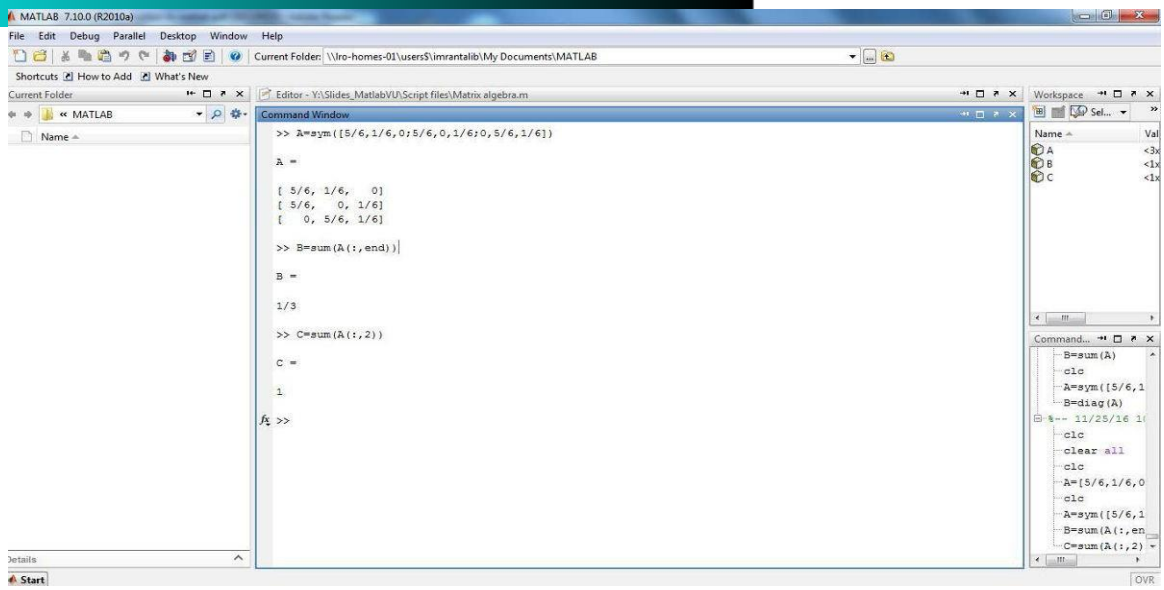


Figure: Display of the sum command with colon operator

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Multiplication of the matrices A and B

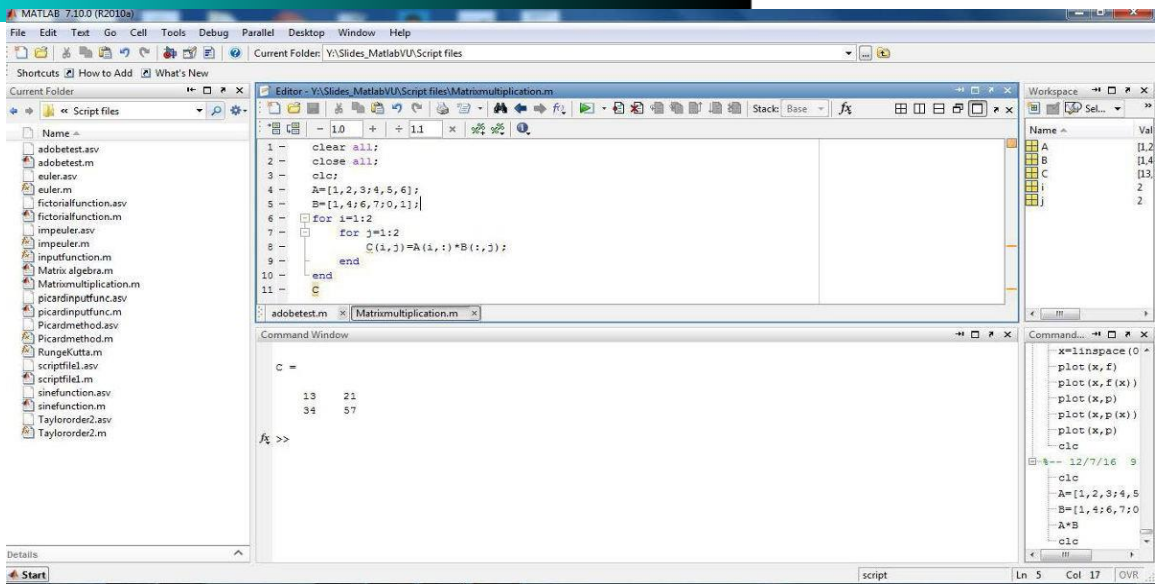


Figure: Matrices multiplication using for loop

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Multiplication of the matrices A and B.

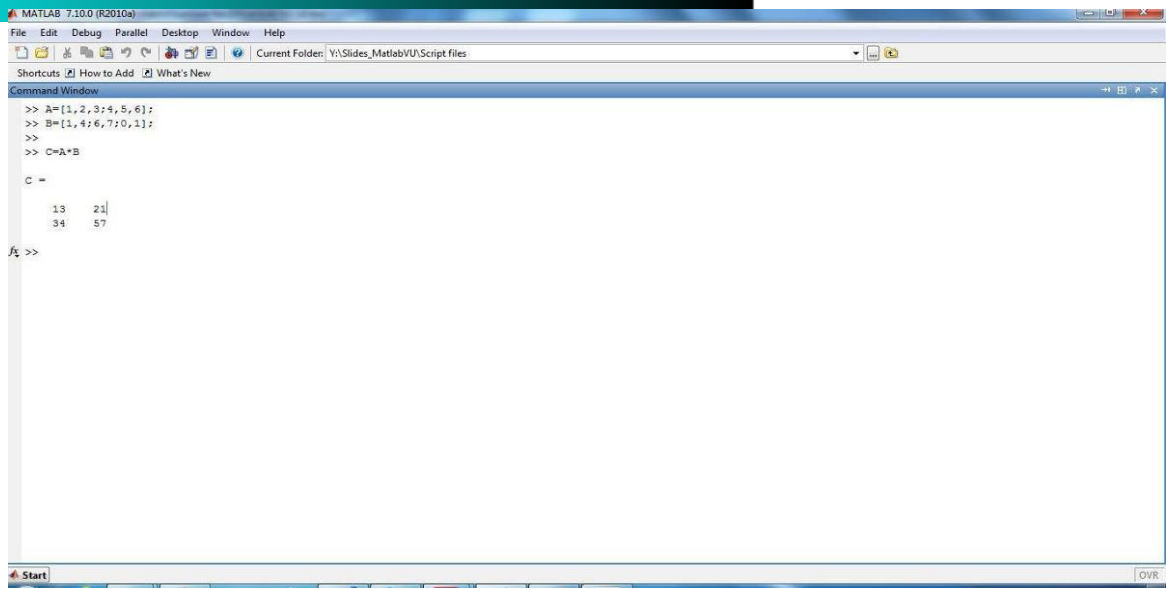


Figure: Matrices multiplication using asterisk operator

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The inverse of a square matrix in MATLAB is determined using the inv command.

For example, the inverse of the matrix A is computed as

» inv(A)

The required output is as under:

ans =

$$\begin{bmatrix} \frac{150}{131} & \frac{36}{131} & -\frac{30}{131} \\ \frac{36}{131} & -\frac{180}{131} & \frac{150}{131} \\ -\frac{30}{131} & \frac{150}{131} & \frac{6}{131} \end{bmatrix}$$

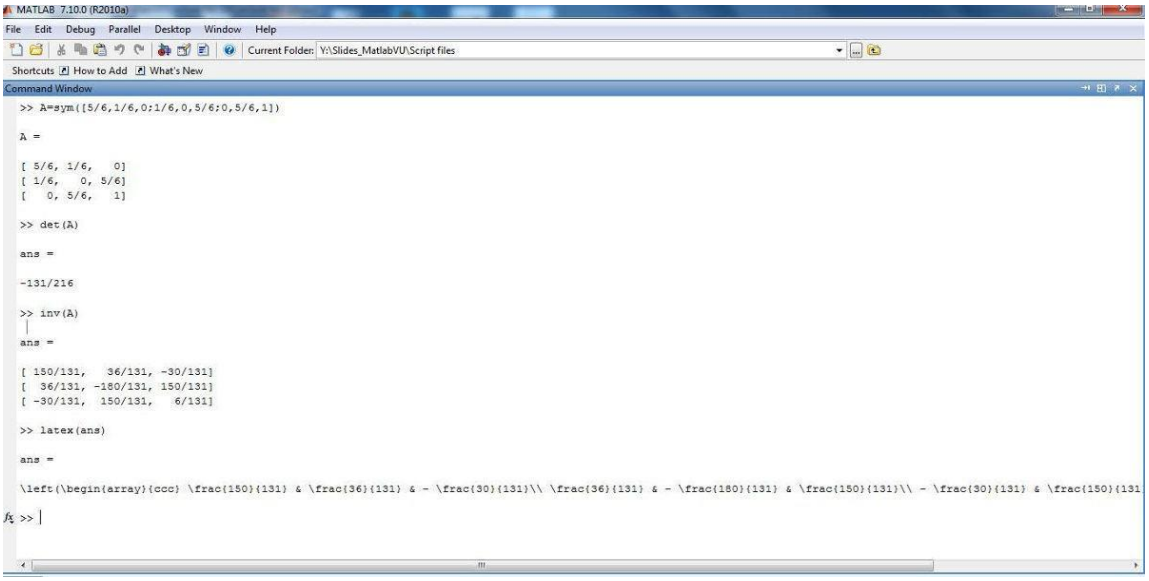


Figure:

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Compute AB using for loop in MATLAB, if

$$A=\begin{bmatrix} \frac{5}{6} & \frac{1}{6} & 0 \\ \frac{1}{6} & 0 & \frac{5}{6} \\ 0 & \frac{5}{6} & 1 \end{bmatrix}$$

$$B=\begin{bmatrix} \frac{5}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{5}{3} \\ 0 & \frac{5}{6} & 1 \end{bmatrix}$$

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